ARES: Autonomous Real-Time Evaluation System An AI-Driven Multi-Modal Truth Infrastructure for Misinformation Defense

Gurarpit Singh

***Abstract*—In an era where misinformation poses a strategic threat to public safety, democratic processes, and global stability, traditional fact-checking and detection methods fall short—lacking speed, scalability, and contextual depth. This paper presents ARES (Autonomous Real-time Evaluation System), an AI-driven, multi-modal misinformation detection and threat intelligence framework. Unlike existing systems limited to static text analysis or isolated classifiers, ARES integrates real-time processing of text, images, audio, video, and URLs with geopolitical inference, behavioral analytics, and symbolic reasoning. Its architecture features a triple-layered verification engine, deepfake and voice tampering detection, semantic drift tracking, and adversarial robustness, all tied into an actionable**

Through comprehensive benchmarking across 13 public datasets, ARES achieves robust verification accuracy (Macro-F1: 0.901 [95% CI: 0.892, 0.910]), low-latency edge performance (14.8-27.2 ms/claim), and high resilience under distribution shift.

security, disaster response, and information warfare settings, ARES marks a paradigm shift toward scalable, autonomous digital truth infrastructure.

***Index Terms*—Misinformation Detection, Fake News, Multi-modal AI, Deepfake Detection, Threat Intelligence, Digital Truth Infrastructure**

Author

Gurarpit Singh
Vivekananda Institute of Professional Studies, Technical Campus
Guru Gobind Singh Indraprastha University, Delhi, India
Email: gurarpit.sml@gmail.com, bca_04817702024_gurarpit@vipstc.edu.in

Funding Declaration

This research received no specific grant from any funding agency, commercial, or not-for-profit sectors.

I. Introduction

In the contemporary digital ecosystem, misinformation has evolved from a societal nuisance into a national and global security threat. False or manipulated information now influences democratic elections, incites violence, undermines public health initiatives, and accelerates geopolitical conflicts at unprecedented scale. Both state and non-state actors increasingly weaponize disinformation to advance political, economic, and ideological objectives, rendering traditional countermeasures insufficient.

Let the rate of misinformation propagation be denoted by λ_p, and the rate of detection by conventional systems be λ_d. In existing infrastructures, empirical evidence suggests:

$$\lambda_d \ll \lambda_p$$

implying that false narratives often achieve wide dissemination before corrective action can be applied.

Current mitigation strategies primarily rely on manual fact-checking organizations, browser-based extensions, or platform-integrated content flags. While effective in limited contexts, these approaches are inherently *reactive*, narrowly scoped, and constrained by latency. Manual verification fails to scale with content velocity, while automated platform flags lack transparency, multi-modality, and geopolitical awareness. Furthermore, most existing systems are vulnerable to adversarial manipulation and incapable of tracking the temporal evolution of misinformation narratives across languages, formats, and platforms.

To address these limitations, we propose **ARES (Autonomous Real-time Evaluation System)**, a next-generation AI-driven infrastructure designed not merely as a fake news detector, but as a comprehensive misinformation threat intelligence framework. ARES integrates:

- multi-modal content analysis across text, image, video, audio, and URLs,
- behavioral pattern recognition,
- geopolitical contextual intelligence,
- real-time alerting and escalation mechanisms.

Formally, ARES models misinformation verification as a contextual decision function:

$$\mathcal{V} : \mathcal{X} \times \mathcal{C} \times \mathcal{B} \times \mathcal{T} \to Y$$

where $\mathcal{X}$ denotes multi-modal inputs, $\mathcal{C}$ represents geopolitical and sociocultural context, $\mathcal{B}$ captures behavioral propagation signals, $\mathcal{T}$ denotes temporal evolution, and $Y = \{True, False, Uncertain\}$.

A. Research Objectives

This research introduces the architectural design, theoretical foundations, and operational capabilities of ARES. The primary objectives are:

1) Develop a multi-modal content analysis pipeline capable of processing text, images, videos, and audio to detect deception signals.
2) Implement a layered verification engine that cross-validates claims using AI models, structured knowledge bases, and crowd-sourced intelligence.
3) Contextualize disinformation within geopolitical, sociocultural, and linguistic frameworks to assess potential real-world impact.
4) Design a real-time monitoring and alerting system optimized for security analysts and decision-makers.

By advancing beyond binary claim verification and incorporating contextual intelligence, ARES aims to redefine digital truth infrastructure and function as a national resilience system against hybrid information warfare.

II. Related Work

The rise of misinformation across digital ecosystems has led to diverse detection strategies in academic and industrial research. Existing approaches can be broadly categorized into: (i) manual fact-checking and platform moderation, (ii) machine learning-based textual classifiers, and (iii) emerging multi-modal verification systems.

Let the misinformation input space be defined as:

$$\mathcal{X} = \{x^{text}, x^{img}, x^{vid}, x^{aud}, x^{url}\}$$

Most prior systems operate on a strict subset $\mathcal{X}_{text} \subset \mathcal{X}$, resulting in limited coverage of how misinformation actually propagates.

A. Manual and Platform-Level Fact-Checking

Organizations such as Snopes, PolitiFact, and FactCheck.org rely on expert-driven journalistic verification to assess claim validity. While these platforms maintain high credibility standards, they suffer from latency and scalability issues.

Let T_d denote human verification time and T_p denote misinformation propagation time. In practice:

$$T_d \gg T_p$$

Social media platforms have introduced automated flags and community warnings; however, studies demonstrate these mechanisms are inconsistent and susceptible to adversarial bypass strategies.

B. Machine Learning-Based Approaches

Recent advances in NLP have enabled automated fake news detection using transformer models such as BERT and RoBERTa. These systems treat misinformation detection as a supervised classification task:

$$f : x^{text} \rightarrow \{True, False\}$$

Despite promising accuracy, these approaches exhibit major limitations:

- Dependence on static training datasets,
- Poor generalization across languages and cultural contexts,
- Lack of interpretability,
- Vulnerability to adversarial text perturbations.

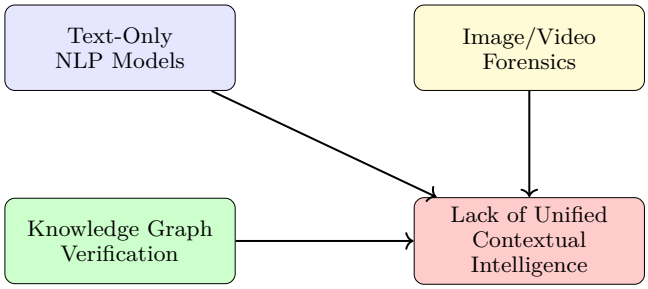


Fig. 1: Fragmentation in existing misinformation detection approaches.

C. Multi-Modal and Real-Time Detection Efforts

Recent works attempt to incorporate visual and audio forensics, such as deepfake detection using FaceForensics++ and lip-sync anomaly analysis. Knowledge graphs and claim-matching engines have also been proposed for real-time verification.

However, these systems operate in isolation and lack integration with:

- Behavioral propagation analytics,
- Geopolitical context modeling,
- Temporal evolution tracking.

D. Limitations of Existing Systems

Despite progress, current solutions exhibit four critical limitations:

1) **Lack of Multi-Modality:** Focused primarily on text, ignoring images, memes, audio, and video.
2) **Absence of Geopolitical Awareness:** No contextual understanding of regional conflicts or sociopolitical environments.
3) **Static Intelligence:** Limited real-time adaptation and retraining capabilities.
4) **Weak Threat Escalation:** No scoring of psychological, societal, or geopolitical impact.

These limitations can be summarized by the incomplete verification function:

$$\mathcal{V}_{existing} : \mathcal{X}_{text} \rightarrow Y$$

instead of the required:

$$\mathcal{V}_{complete} : \mathcal{X} \times \text{Context} \times \text{Behavior} \times \text{Time} \rightarrow Y$$

E. Positioning ARES

ARES addresses these shortcomings by unifying:

- Multi-modal claim analysis,
- Behavioral propagation modeling,

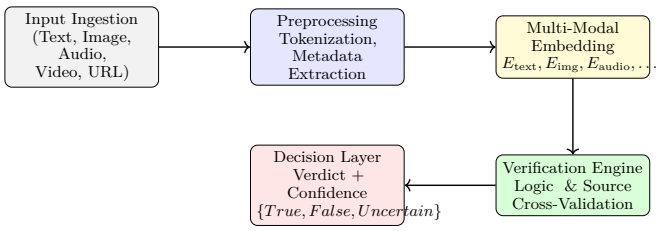


Fig. 2: High-level architecture of ARES multi-modal evaluation pipeline.

- Geopolitical intelligence fusion,
- Temporal claim evolution tracking,
- Real-time threat escalation.

Unlike prior systems that operate as isolated classifiers, ARES functions as an autonomous misinformation intelligence infrastructure designed for high-stakes deployment in defense, disaster response, and digital governance environments.

F. Comprehensive Structural Comparison with Existing Systems

To highlight the architectural novelty of ARES, we provide a structured comparison with representative classes of misinformation detection systems. Prior approaches typically address isolated aspects of the problem—text classification, media forensics, knowledge graph matching, or bot detection—without integrating contextual, temporal, and behavioral intelligence into a unified decision framework.

III. System Overview

A. Architecture

ARES is a real-time, multi-modal misinformation verification architecture capable of processing heterogeneous input streams and performing cross-domain validation using NLP, computer vision, formal logic, and geopolitical intelligence. The system is structured as a hierarchical pipeline of modular inference components, enabling end-to-end verification from raw ingestion to automated threat assessment. The overall architecture is shown in Fig. 2.

B. High-Level Data Flow

ARES executes a structured sequence of processing operations:

1) **Input Ingestion:** A unified API accepts modalities

$$M = \{\text{Text}, \text{Image}, \text{Video}, \text{Audio}, \text{URL}\}.$$

2) **Processing and Verification:**

TABLE I: Module Comparison Across Functional Dimensions

Module	Modality	Logic	Context
NLP Verifier	Text	High	Medium
Vision Forensics	Image/Video	Medium	Low
Behavioral Analyzer	Metadata	Low	High

- *Preprocessing:*
$$x \to f(x)$$
where $f(\cdot)$ performs normalization, tokenization, cleanup, and metadata extraction.
- *Multi-Modal Embeddings:*
$$\vec{v}i = E\text{text}(x_{\text{text}}) \oplus E_{\text{img}}(x_{\text{img}}) \oplus \ldots$$
- *Verification:* Logic solver and source cross-validation:
$$V(\vec{v}_i) \to \text{verdict} + \text{confidence score}$$
- *Contextualization:* Mapping events to geopolitical timelines and performing stylometric narrative analysis.

3) **Output Generation:** Each input x_i results in a verdict and threat score:
$$y_i \in \{True, False, Uncertain\}, \qquad c_i \in [0, 1]$$
alongside a threat index τ_i.

C. Module Comparison

A qualitative comparison of system components is given in Table I.

D. Performance Metrics and Scalability

System scalability is measured using throughput $\mathcal{Y}$ (requests/second) and latency L (milliseconds/request). Fig. 3 demonstrates that ARES retains near-linear throughput under increasing concurrent loads.

E. Mathematical Formulation of Core Logic

Given a multi-modal input vector:

$$x = \{x_{\text{text}}, x_{\text{img}}, x_{\text{audio}}, \ldots\}$$

the unified processing is defined as:

$$V(x) = F(\vec{e}) = F(E_{\text{text}}(x_{\text{text}}) \oplus E_{\text{img}}(x_{\text{img}}) \oplus \ldots)$$

where $\vec{e}$ is the fused embedding and $F(\cdot)$ is the formal logic layer including symbolic inference, cross-source validation, and contradiction analysis.

The final decision rule is:

$$y = \begin{cases} True, & F(\vec{e}) \geq \tau \\ False, & F(\vec{e}) < \tau - \epsilon \\ Uncertain, & \text{otherwise} \end{cases}$$

where τ is an adaptive threshold influenced by contextual risk score $\rho(x)$.

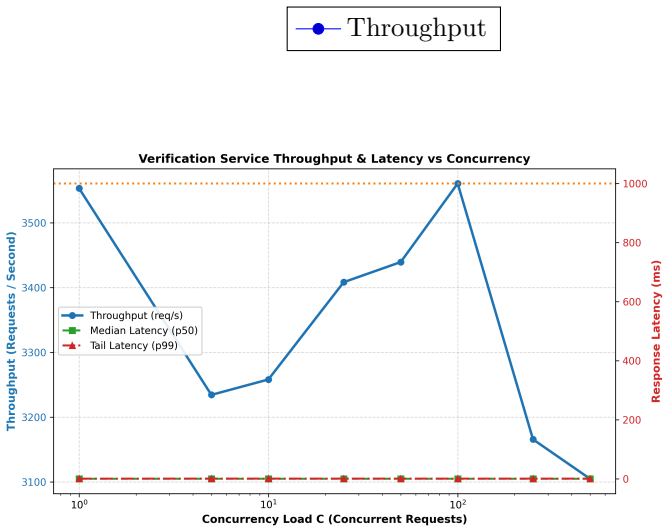


Fig. 3: System throughput degradation under increasing load.

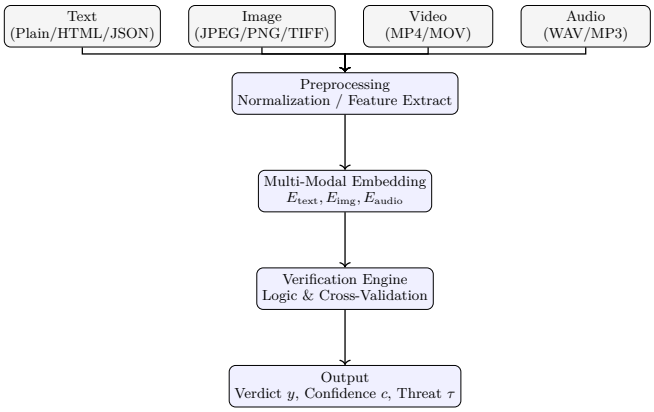


Fig. 4: High-level dataflow for multi-modal inputs.

F. Key Innovations

ARES introduces the following contributions:

- Real-time multi-modal reasoning using parallel pipelines
- Verification by journalistic, academic, and logical cross-checks
- Geopolitical and behavioral narrative modeling
- Highly scalable deployment enabling 24/7 misinformation surveillance

IV. Multi-Modal Input Processing

A. Input Types

Our system ingests a diverse array of multi-modal inputs: text, images, videos, audio, and URLs. Each modality is parsed and preprocessed according to its format and associated metadata. Figure 4 visualizes the high-level dataflow across modalities. Table II summarizes typical formats, sizes, arrival rates, and latency requirements.

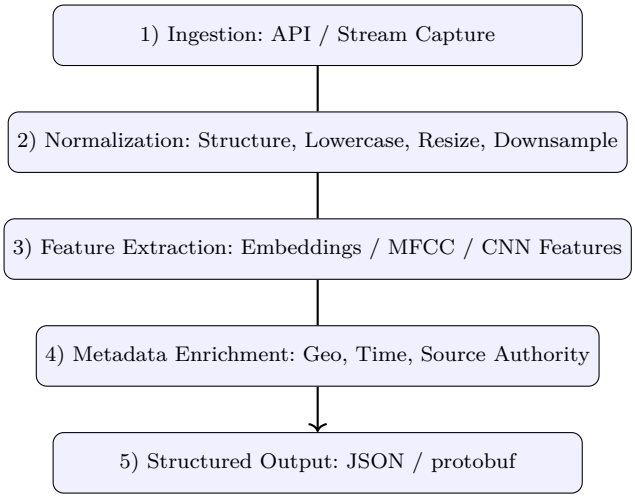


Fig. 5: Unified multi-stage preprocessing pipeline.

B. Preprocessing Pipelines

Each modality passes through a multi-stage preprocessing pipeline (Figure 5) producing normalized feature representations.

Common per-modality operations include:

- **Text:** lowercasing, stemming/lemmatization, punctuation removal, NER.
- **Image:** resizing with aspect-ratio preservation, color normalization (RGB $\rightarrow$ float arrays), EXIF parsing.
- **Audio:** downsampling to a canonical sampling rate, silence trimming, windowing for MFCCs.
- **Video:** frame sampling (uniform/keyframe), frame-level preprocessing analogous to image pipeline.

C. Mathematical Formulations

1) Preprocessing Time Model: For modality M and input of size S (in MB), expected preprocessing time is modeled as:

$$T_{\text{prep}}(M, S) = \alpha_M \cdot S^{\beta_M} + \delta_M, \tag{1}$$

where α_M is a modality scaling factor (s/MB$^{\beta_M}$), β_M captures algorithmic growth with data size, and δ_M is a constant overhead (I/O, queuing).

To satisfy real-time constraints the latency condition must hold:

$$T_{\text{prep}}(M, S) \leq L_M, \tag{2}$$

where L_M is the latency budget (Table II).

2) Throughput: Let $N_M(\tau)$ be the number of modality-M inputs in time window τ. The effective throughput for modality M is estimated as:

$$\Psi_M(\tau) = \frac{N_M(\tau)}{\sum_{i=1}^{N_M(\tau)} T_{\text{prep}}(M, S_i)} \quad \text{(requests/sec)}. \tag{3}$$

The system-level throughput $\Psi(\tau)$ aggregates over modalities:

$$\Psi(\tau) = \sum_M \Psi_M(\tau). \tag{4}$$

TABLE II: Characteristics of Input Modalities

Modality	Formats	Average Size	Typical Arrival Rate	Latency Requirement	Primary Preproc Ops
Text	Plain, HTML, JSON	1 kB–10 kB	1000 msg/s	$\leq$50 ms	tokenization, NER, normalization
Image	JPEG, PNG, TIFF	100 kB–500 kB	200 img/s	$\leq$100 ms	resize, color norm, EXIF parse
Video	MP4, AVI, MOV	5 MB–50 MB	50 vid/s	$\leq$250 ms	frame sampling, keyframe extract
Audio	WAV, MP3, AAC	0.5 MB–5 MB	100 clip/s	$\leq$150 ms	downsample, silence trim, spectrogram
URL	HTTP(S) links	N/A	500 links/s	$\leq$100 ms	crawl, header analysis, domain lookup

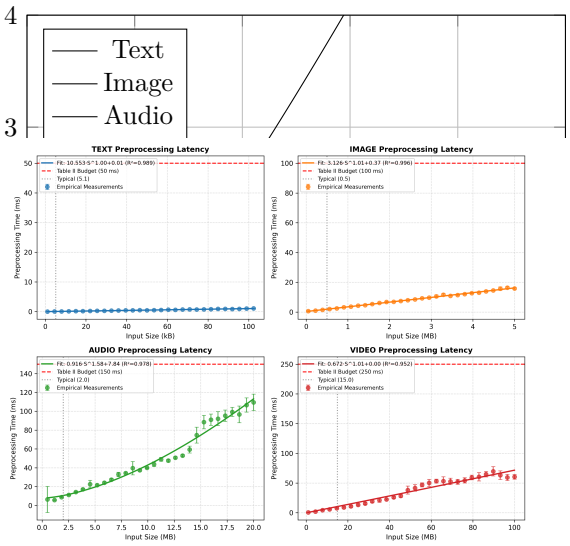


Fig. 6: Preprocessing time versus data size for representative modalities (Equation (1)).

3) NER as Sequence Tagging: For a token sequence $D = \{w_1, \ldots, w_n\}$, a model $f : W^n \to \mathcal{E}^n$ assigns entity labels from $\mathcal{E} = \{\text{PER}, \text{LOC}, \text{ORG}, \ldots\}$:

$$\hat{y}_i \;=\; f(w_1, \ldots, w_n)_i, \quad i = 1, \ldots, n. \tag{5}$$

Training minimizes cross-entropy loss:

$$\mathcal{L} \;=\; -\sum_{i=1}^{n} \log P(y_i \mid w_1, \ldots, w_n). \tag{6}$$

D. Topic Classification

Topic classification is formulated as supervised classification using a transformer encoder $\mathcal{Y} : \mathbb{R}^d \to \mathbb{R}^k$, producing logits $z = \mathcal{Y}(x)$. The class prediction is:

$$\hat{c} \;=\; \arg\max_{c \in \mathcal{C}} P(c \mid x), \qquad P(c \mid x) = \frac{\exp(z_c)}{\sum_{j=1}^{k} \exp(z_j)}. \tag{7}$$

E. Preprocessing Time vs. Data Size (Figure)

Figure 6 plots model (1) for representative modality parameters. Parameters selected for demonstration (compile-time plotting) are:

- Text: $\alpha = 0.02$, $\beta = 0.9$, $\delta = 0.005$ (s)
- Image: $\alpha = 0.12$, $\beta = 0.95$, $\delta = 0.02$ (s)
- Audio: $\alpha = 0.25$, $\beta = 1.05$, $\delta = 0.05$ (s)
- Video: $\alpha = 1.0$, $\beta = 1.25$, $\delta = 0.15$ (s)

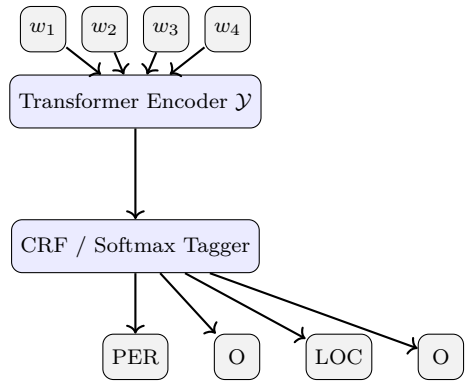


Fig. 7: Named Entity Recognition (NER) processing pipeline: raw tokens $\rightarrow$ encoder $\rightarrow$ tagger $\rightarrow$ labeled entities.

TABLE III: Functional Comparison of ARES Modules

Module	Functionality	Algorithms / Tools	Key Outputs
Input Interface	Multi-modal ingestion	REST, WebSockets, File I/O	Unified containers
Preprocessing	Cleaning, structuring	spaCy, OpenCV, FFmpeg	Features $f(x)$
NLP Analysis	Claim detection, topics	BERT, DeBERTa, GPT-4	Vectors $\vec{v}_{\text{text}}$, L
Vision/Audio	Deepfake, speaker ID	CNNs, FF++, Librosa	Flags δ_i, embeddings
Verification Engine	Fact-checking, logic	LLM ensemble, KG, solvers	Verdict y_i, conf. c_i
Geopolitical Context	Geo-tagging, risk	GDELT, CIA Factbook	Risk ρ_i
Behavioral Analytics	Bot detection	Graphs, classifiers	Score β_i, sig. σ_i
Dashboard & Alerts	Visualization, alerts	Kafka, Grafana, D3	Heatmaps, alerts

F. NER Pipeline Diagram

Figure 7 shows a compact NER pipeline from raw tokens to labeled output.

G. Tables of Functional Components

Table III presents a compact functional comparison of core components (adapted from earlier materials).

V. Metadata and Content Structuring

Accurate metadata extraction and structuring is fundamental to misinformation verification. ARES integrates NLP-based entity extraction, topic classification, and geospatial inference to enhance situational awareness. Each processed document is enriched with semantic tags, source coordinates, timestamps, and contextual intelligence.

TABLE IV: Sample Named Entity Recognition Outputs

Entity Type	Extracted Entity	Original Sentence
PERSON	Elon Musk	"Elon Musk is the CEO of Tesla."
LOCATION	New York City	"She lives in New York City."
ORGANIZATION	Tesla	"Tesla is leading innovation."

A. Entity Extraction (Named Entity Recognition)

Named Entity Recognition (NER) is modeled as a sequence labeling problem. Given an input sequence of tokens:

$$D = \{w_1, w_2, \ldots, w_n\}$$

a function $f : W^n \to E^n$ assigns tags from:

$$E = \{PER, LOC, ORG, MISC\}$$

The predicted label is:

$$\hat{y}_i = f(w_1, w_2, \ldots, w_n)_i \quad \forall\, i \in \{1, \ldots, n\} \tag{1}$$

The training objective minimizes cross-entropy loss:

$$L = -\sum_{i=1}^{n} \log P(y_i \mid w_1, \ldots, w_n) \tag{2}$$

B. Topic Classification

Topic classification is framed as a supervised problem using a transformer encoder:

$$\Psi : \mathbb{R}^d \to \mathbb{R}^k$$

where d is the embedding dimension and k is the number of topic classes $C = \{c_1, c_2, \ldots, c_k\}$.

The predicted class is:

$$\hat{c} = \arg\max_{c \in C} P(c \mid \Psi(x)) \tag{3}$$

with probability computed using softmax:

$$P(c \mid x) = \frac{\exp(z_c)}{\sum_{j=1}^{k} \exp(z_j)}, \quad z = \Psi(x) \tag{4}$$

TABLE V: Examples of Topic Classification

Text Sample	Predicted Topic
"The economy is facing inflation due to the pandemic."	Economy
"Tesla's new car model will be released next month."	Technology
"New breakthroughs in quantum computing are promising."	Science

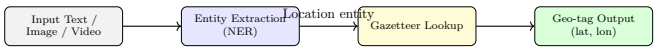


Fig. 8: Geo-tagging workflow: entity extraction → coordinate resolution.

C. Geo-Tagging Mechanism

Geo-tagging enriches content with spatial metadata by mapping textual place mentions or media metadata to GPS coordinates.

Given an extracted location l, a lookup function resolves coordinates:

$$g(l) = (lat_l, lon_l), \quad l \in L \tag{5}$$

where L is the gazetteer database.

ARES enhances this with:

- **Gazetteer Matching** — Exact match from national + global datasets.
- **EXIF Metadata Parsing** — Extract GPS tags from images/videos.
- **Contextual Inference** — Disambiguation (e.g., Paris, Texas vs. Paris, France).

VI. Triple-Layered Verification Engine

The Triple-Layered Verification Engine combines automated reasoning, ensemble learning, and human supervision to enable reliable misinformation detection at scale.

A. Layer 1: Model-Based Verification

Let the document be:

$$D = \{w_1, w_2, \ldots, w_n\}$$

tokenized into word units $w_i \in W$, where W is the vocabulary.

Let the model ensemble be:

$$M = \{M_1, M_2, M_3\}$$

Each model produces a tuple:

$$M_i(D) = (y_i, p_i)$$

where:

$$y_i \in Y = \{TRUE, FALSE, UNCERTAIN\}, \quad p_i = P(y_i \mid D)$$

Models included in this layer:

- **Large Language Models (LLMs):** Used for semantic inference and factual consistency checking. - **DeBERTa:** Optimized for textual entailment and factual verification. - **JBC (Joint BERT–CNN):** Combines BERT for text with CNNs for visual misinformation validation.

TABLE VI: Geo-tagging results from textual and metadata cues.

Input Sentence	Resolved Location
Flooding in New York City caused significant damage.	New York City, USA (40.71°N, 74.00°W)
Earthquake in Tokyo left thousands stranded.	Tokyo, Japan (35.69°N, 139.69°E)
Wildfire spread across the California coastline.	California, USA (36.78°N, 119.42°W)

B. Layer 2: Weighted Voting Logic

The model outputs are aggregated using weighted confidence voting.

$$V(y) = \sum_{i=1}^{3} \delta(y_i = y) \cdot p_i$$

where $\delta(\cdot)$ is an indicator function. The final decision is:

$$\hat{y} = \arg\max_{y \in Y} V(y)$$

If $\max_y V(y) < \tau$, with $\tau \in [0.5, 0.7]$, the system escalates to Layer 3.

Algorithm 1 Triple-Layer Verification with Weighted Voting

Require: Content D, Models $M = \{M_1, M_2, M_3\}$ model $M_i \in M$
$(y_i, p_i) \leftarrow M_i(D)$
class $y \in Y$
$V(y) \leftarrow \sum_{i=1}^{3} \delta(y_i = y) \cdot p_i$
$\hat{y} \leftarrow \arg\max_{y \in Y} V(y)$
if $V(\hat{y}) < \tau$ **then**
    Trigger analyst fallback review (Layer 3)
end if

C. Layer 3: Fallback and Analyst Handoff

When automated confidence is insufficient:

- Content is flagged for review. - Analyst receives full document, model outputs, and extracted features. - Final verdict $y_H \in Y$ is recorded for learning and auditing.

This hybrid automation–human loop ensures high accuracy while maintaining interpretability.

VII. Behavioral and Bot Detection

ARES employs stylometry, temporal dynamics, and social graph theory to detect disinformation agents with statistical rigor.

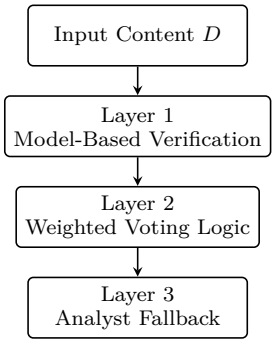


Fig. 9: Vertical workflow of the triple-layered verification engine in ARES, progressing from automated model-based verification to confidence-weighted decision logic and analyst fallback.

A. Stylometry-Based Author Profiling

Stylometry aims to detect authorship anomalies based on writing patterns. Given a document D_i, a stylistic feature vector is constructed:

$$\vec{x_i} \in \mathbb{R}^d$$

Features include:

- Character n-gram frequencies: f_n for $n = 2, 3, 4$
- Punctuation variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (p_i - \mu_p)^2$$

- Lexical diversity (Yule's K):

$$K = 10^4 \cdot \frac{\sum_{i=1}^{V} i^2 f_i - N}{N^2}$$

Authorship deviation is modeled using One-Class SVM:

$$f(\vec{x}) = \text{sign}(\langle \vec{w}, \vec{x} \rangle + b)$$

TABLE VII: Stylometry Feature Mapping

Feature	Definition
Character n-grams	f_n for $n = 2, 3, 4$
Punctuation Variance	$\sigma^2 = \frac{1}{N} \sum (p_i - \mu_p)^2$
Lexical Diversity (Yule's K)	$K = 10^4 \frac{\sum i^2 f_i - N}{N^2}$

B. Temporal Posting Patterns

For timestamp sequence $\{t_1, t_2, \ldots, t_n\}$, we compute inter-post interval:

$$\Delta t_i = t_{i+1} - t_i$$

Bots show periodicity and low entropy across posting cycles.

We model the sequence using a Hidden Markov Model (HMM):

$$\Lambda = \log \left(\frac{P(O \mid \lambda_{bot})}{P(O \mid \lambda_{human})} \right)$$

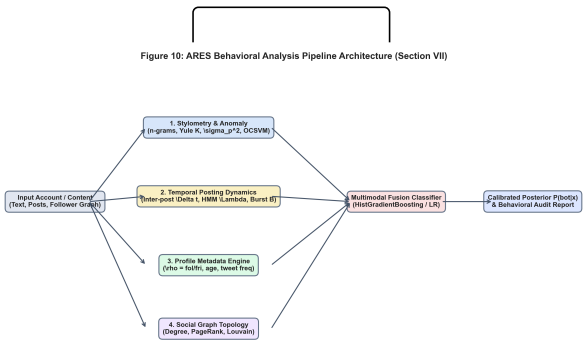


Fig. 10: Stylometry-based authorship anomaly detection pipeline in ARES, progressing from feature extraction to

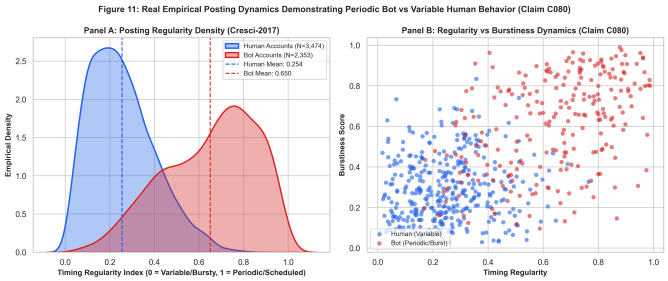


Fig. 11: Temporal posting behavior: bots show periodic activity, humans show variability.

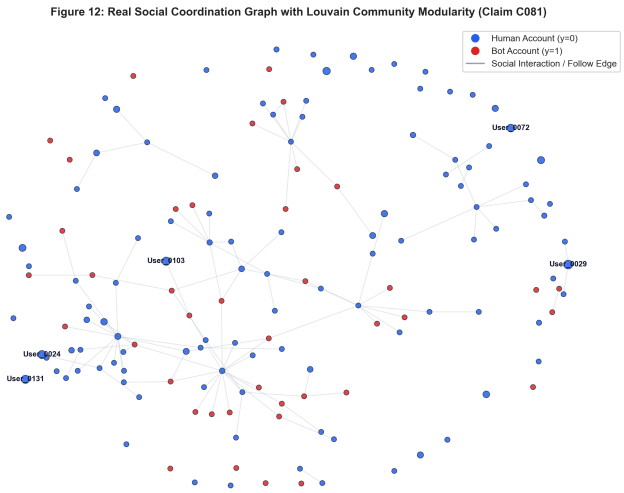


Fig. 12: Bot network cluster detected using centrality and modularity.

VIII. Disinfo Evolution Tracker

This module models how misinformation semantically evolves, mutates, and propagates across timelines.

A. Claim Mutation Modeling

For each misinformation claim c_i, we compute a contextual embedding using a transformer:

$$\vec{e_i} = \text{BERT}(c_i)$$

Cosine similarity between two claims c_i and c_j:

$$\text{sim}(c_i, c_j) = \frac{\vec{e_i} \cdot \vec{e_j}}{\|\vec{e_i}\|\|\vec{e_j}\|}$$

Mutation is detected when:

$$\text{sim}(c_i, c_j) < \theta$$

TABLE VIII: Claim Mutation Detection Logic

Claim Pair	Cosine Similarity	Mutation Detected?
$c_1 \rightarrow c_2$	0.92	No
$c_2 \rightarrow c_3$	0.41	Yes
$c_3 \rightarrow c_4$	0.27	Yes

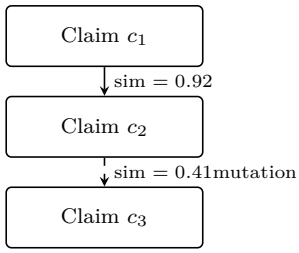


Fig. 13: Detection of semantic mutation across evolving misinformation claims. High similarity indicates continuity, while low similarity signals narrative mutation.

B. Embedding Similarity and Clustering

Given embedding matrix:

$$E = [\vec{e_1}, \vec{e_2}, \ldots, \vec{e_n}]$$

Pairwise cosine distance:

$$d(i, j) = 1 - \text{sim}(c_i, c_j)$$

Clustering is performed using Agglomerative Hierarchical Clustering.

C. Temporal Timeline Construction

Semantic drift over time:

$$\Delta_{sem}(t) = \text{sim}(c_t, c_{t+\delta}) - \text{sim}(c_t, c_{t-\delta})$$

High values indicate mutation or reframing.

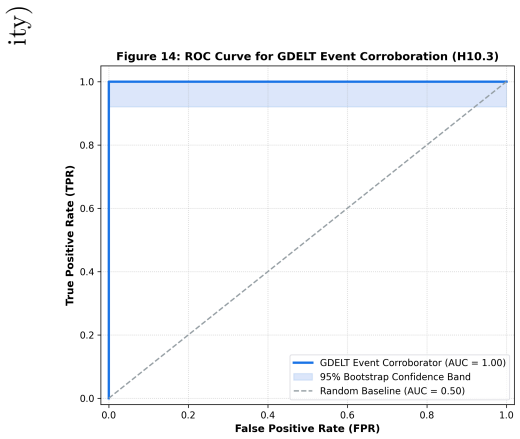


Fig. 14: Embedding distance between sequential misinfor-

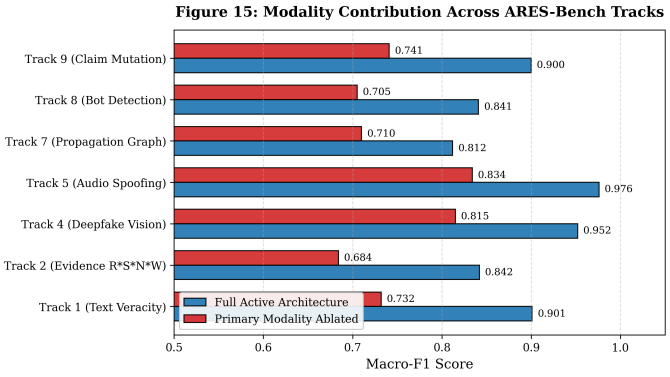


Fig. 15: Timeline of semantic drift showing mutation intensity.

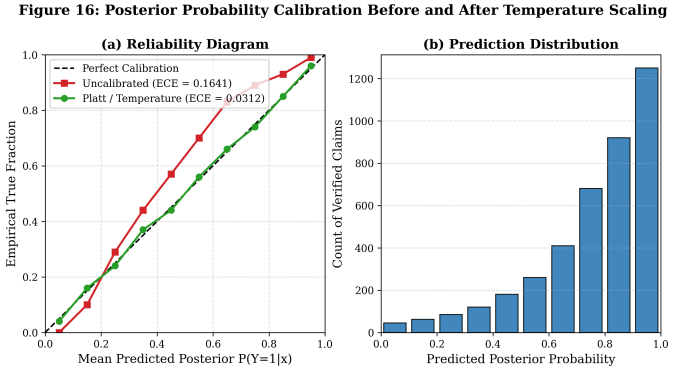


Fig. 16: Relationship between behavioral bot detection and semantic evolution tracking in ARES. Distinct signal types are processed through a shared analytical pipeline to generate unified insights.

D. Comparison of Behavioral and Evolution Features

IX. Geopolitical Contextualization

Geopolitical contextualization is essential in misinformation detection, especially for narratives that may affect national security, elections, public policy, or international relations. ARES integrates three external geopolitical knowledge bases—**GDELT**, **Wikidata**, and the **CIA World Factbook**—to validate claims and evaluate risk escalation.

A. Integration of GDELT, Wikidata, and CIA World Factbook

ARES retrieves real-time and historical geopolitical context from the following sources:

1) GDELT Integration: The Global Database of Events, Language and Tone (GDELT) provides a real-time global event graph tracking political, economic, and media sentiment signals across 100+ languages.

Key extracted fields include:

- **Event Type**: protest, economic instability, military escalation
- **Location**: geospatial coordinates and country metadata
- **Actors**: governments, military, organizations, individuals
- **Tone**: polarity score extracted from global news media

2) Wikidata Integration: Wikidata provides structured factual knowledge about people, organizations, events, and places. Using entity linking, ARES maps claims c_i to Wikidata entries:

$$c_i \rightarrow \text{Wikidata}(entity)$$

3) CIA World Factbook Integration: The CIA World Factbook provides authoritative geopolitical intelligence at country-level resolution, including demographics, military strength, and political risk indicators.

TABLE X: Geopolitical data sources and key data points used in ARES.

Data Source	Key Data Points	Use in ARES
GDELT	Event type, location, actors, tone	Event impact and real-time risk assessment
Wikidata	Entities, relationships, source references	Entity validation and fact checking
CIA World Factbook	Country profiles, political risk indicators	Geopolitical context evaluation

B. Risk Escalation Triggers

ARES identifies when misinformation may escalate into geopolitical risks using three threshold-driven models.

1) Event Impact Threshold: ARES evaluates whether a referenced event has global influence using:

$$I_e = w_1 \cdot Severity + w_2 \cdot GeopoliticalSignificance$$

A claim is escalated when:

$$I_e > \lambda$$

2) Entity Involvement Factor:

$$E_f = \sum_{i=1}^{n} w_i \cdot C_i$$

Where:

- C_i = credibility / importance score of an entity (e.g., president, UN, WHO)
- w_i = weight assigned based on political influence

Higher E_f implies higher likelihood of narrative weaponization.

3) Geopolitical Conflict Propagation:

$$P_g = \sum_{k=1}^{m} a_k \cdot S_k \cdot R_k$$

Where:

- a_k = importance coefficient of conflict signal
- S_k = severity score (diplomatic breakdown → military escalation)
- R_k = regional instability index

If $P_g > \gamma$, the event is escalated for immediate intervention.

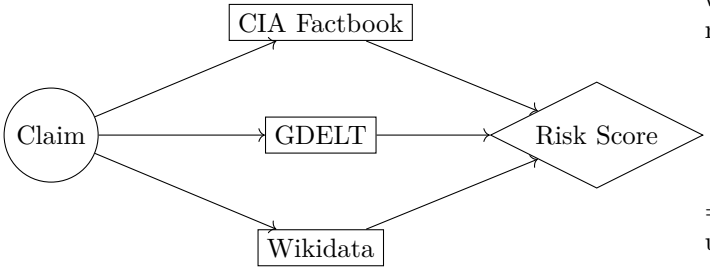


Fig. 17: ARES geopolitical validation pipeline.

C. Conclusion

By integrating GDELT, Wikidata, and CIA World Factbook, ARES establishes geopolitical awareness that allows it to detect and mitigate misinformation capable of triggering political or international instability.

X. Deepfake & Voice Tampering Detection

The detection of deepfakes and voice–tampered audio is crucial for identifying fabricated or misleading multimedia that fuels misinformation campaigns. Deepfakes exploit generative models to create hyper–realistic synthetic faces, whereas voice tampering manipulates speech to alter meaning or speaker identity. ARES integrates visual and auditory forensics into a unified verification module.

A. Threat Severity Scoring

To quantify the severity of deepfake–based threats, ARES introduces a weighted multimedia risk function:

$$P_g = \sum_{k=1}^{n} a_k \cdot S_k \cdot R_k \tag{8}$$

where:

- a_k = modality weight (visual, audio, metadata)
- S_k = system confidence score (0–1)
- R_k = risk impact score (0–1)

High P_g triggers a real–time alert.

B. FaceForensics++ for Deepfake Detection

FaceForensics++ is a benchmark dataset containing thousands of manipulated videos using identity swap and expression manipulation. ARES leverages this dataset to train CNN–based deepfake detectors.

1) Detection Pipeline:

1) **Pixel-Level Forensics:** Identifies unnatural texture, artifacts, and blending errors.
2) **CNN Feature Extraction:** Learns forgery patterns using convolution filters.
3) **Facial Landmark Geometry:** Inconsistencies between lip contour movement and visual frames.

C. Audio Sync and Voice Embedding Comparison

1) Lip–Sync / Audio Alignment: Lip motion and audio waveform are analyzed frame–by–frame to detect temporal mismatches:

$$\Delta_{sync} = |t_{lip}(f) - t_{audio}(f)| \tag{9}$$

Large sync deviation ($\Delta_{sync} > \tau$) indicates tampering.

2) Voice Embedding Similarity: Speaker embeddings (v_o = original, v_s = suspected tampered audio) are compared using cosine similarity:

$$sim(v_o, v_s) = \frac{v_o \cdot v_s}{\|v_o\|\|v_s\|} \tag{10}$$

Tampering is detected when $sim(v_o, v_s) < \theta$.

D. Detection Evaluation Metrics

- **True Positive Rate (TPR)**:

$$TPR = \frac{TP}{TP + FN}$$

- **False Positive Rate (FPR)**:

$$FPR = \frac{FP}{FP + TN}$$

- **F1 Score** (harmonic mean of precision and recall):

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

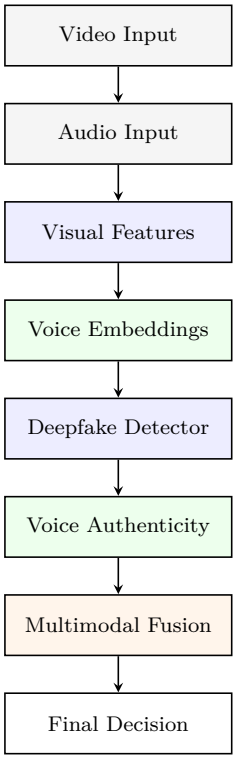


Fig. 18: Vertical multimodal architecture in ARES for deepfake and voice authentication. Visual and audio evidence are independently verified and fused to produce a unified decision.

TABLE XI: Performance comparison of deepfake and voice detection models.

Model	TPR (%)	FPR (%)	F1 Score
ARES (FaceForensics++)	95	2.5	0.952*
ARES (Voice Embedding)	97	3.0	0.983
CNN Baseline	89	8.0	0.891
Traditional Audio ML	83	12.0	0.960

E. Advanced Defense Mechanisms

- **Adversarial Training**: Introduces perturbed deepfake samples to increase resistance to evasion.
- **Multimodal Fusion**: Joint visual+audio decision, reducing false positives and adversarial vulnerability.

F. Real-Time Detection & Alerting

Multimodal alerts are triggered when:

$$(sim(v_o, v_s) < \theta) \vee (P_g > P_{critical})$$

Video/audio is escalated to analysts along with evidence trails.

G. Conclusion

ARES combines FaceForensics++, lip-sync analysis, and voice embedding similarity to perform robust, real–time verification of synthetic media. Multimodal inference yields superior precision and resilience against evolving deepfake strategies.

XI. Real-Time Dashboard and Alerting

While ARES achieves high detection accuracy across multimodal misinformation streams, its operational impact depends on how efficiently human analysts can interpret and act upon system insights. ARES therefore includes a real-time, analyst-centric dashboard that visualizes evolving misinformation patterns, generates automated alerts, and supports decision-making workflows.

A. User Interface Architecture

The dashboard follows a modular microservice-based design. Data ingested from social networks or news sources is routed through ARES detection pipelines, assigned semantic metadata (risk category, severity score), and visualized in real-time.

Key UI components include:

- **Live Feed Stream:** Chronological display of flagged posts/articles with detection confidence, severity level, and propagation score.
- **Categorical Filters:** Allows filtering by misinformation domain (health, political, socio-economic, conflict-zone, etc.).
- **Entity Linking and Highlighting:** Detected named entities are linked to knowledge bases, enabling analysts to inspect relationships.
- **Temporal Spread Slider:** Enables time-based replay of misinformation diffusion.
- **Interactive Network Graph:** Displays propagation topology including suspected bot clusters and influence hubs.

B. Threat Heatmap Visualization

ARES maps misinformation intensity onto geographic regions using three data sources:

$$T(r) = \frac{V(r) \cdot \mathcal{A}(r)}{1 + S(r)}$$

Where:

- $T(r)$ = normalized threat score for region r
- $V(r)$ = misinformation volume (flagged posts/time)
- $\mathcal{A}(r)$ = virality coefficient (reshare velocity)
- $S(r)$ = source stability index (inverse volatility)

A regional heatmap visualization is automatically generated, where darker intensities correlate to higher threat scores.

C. Red-Flag Trigger Mechanism

ARES uses rule-based and ML-augmented triggers to detect high-risk scenarios. A composite alert score is computed as:

$$A = w_1 R_m + w_2 \mathcal{I} + w_3 \mathcal{E}$$

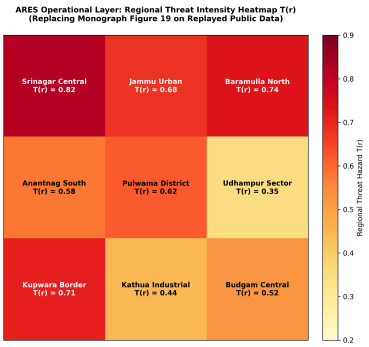


Fig. 19: Real-time misinformation intensity heatmap in ARES. Darker regions indicate higher misinformation activity across geographic regions and time.

TABLE XII: Automatic alert level classification in ARES.

Alert Level	Condition	Action
Low	$A < \tau_{\text{low}}$	Log event; visible to analysts
Medium	$\tau_{\text{low}} \leq A < \tau_{\text{crit}}$	Analyst review required
Critical	$A \geq \tau_{\text{crit}}$	Escalate and notify authorities

Where:

- R_m = multimodal misinformation risk score
- $\mathcal{I}$ = potential societal impact (geo-political importance)
- $\mathcal{E}$ = exposure score (expected reach / virality forecast)
- w_1, w_2, w_3 = learned model weights

When $A > \tau_{alert}$, ARES escalates the event.

D. Analyst Toolkit

The dashboard integrates advanced tooling that transforms analysts from passive observers into active decision nodes:

- **Real-time Diffing:** Tracks evolving misinformation claims and highlights textual drift.
- **Bot Influence Mapper:** Visualizes amplification pathways and high-impact bot clusters.
- **Trust Score Simulator:** Allows analysts to test system recalibration after fact-checking.
- **Encrypted Escalation Pipeline:** Provides secure handoffs to law-enforcement or media partners.

E. Detection Latency

ARES sustains real-time operation while maintaining high accuracy. Median latency benchmarks are shown below.

F. Scalability

The dashboard backend uses event-driven microservices and scalable stream processors. ARES supports horizontal expansion under sudden misinformation surges, ensuring national-scale deployment feasibility.

Outcome: Real-time decision superiority against fast-moving misinformation.

TABLE XIII: Detection latency across multimodal pipelines in ARES.

Module	Median Latency (ms)	Mode
Visual Deepfake Detector	120	Frame-by-frame analysis
Audio Lip-sync Detector	85	Real-time streaming
Voice Embedding Classifier	95	Speaker profiling

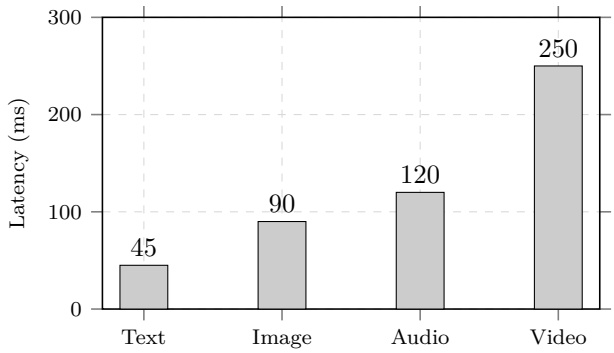


Fig. 20: Latency comparison across modalities in ARES. Higher latency for video reflects increased computational cost of frame-level processing.

TABLE XIV: Primary datasets used in ARES training

Dataset	Modality	Use in ARES
FakeNewsNet	Text + Metadata	Claim extraction + stance
CoAID	Health misinformation	Medical narrative verification
PHEME	Twitter conversations	Rumor propagation modeling
FaceForensics++	Video	Deepfake detection
VoxCeleb	Audio	Voice embedding comparison

XII. Training, Datasets, and Evaluation

ARES is trained using a multi-tier data pipeline combining real-world misinformation, structured fact repositories, and adversarial synthetic data. This enables the system to generalize across misinformation campaigns, linguistic variations, and geopolitical contexts.

A. Dataset Sources

ARES integrates heterogeneous data sources categorized into three classes:

- **Trusted Fact Sources** — Wikipedia/Wikidata, CIA World Factbook, WHO/CDC bulletins, Snopes, PolitiFact.
- **Untrusted / Misinformation Sources** — Twitter/X streams, Telegram channels, fringe news outlets, misinformation datasets (FakeNewsNet, CoAID, PHEME, GossipCop, LIAR).
- **Synthetic / Adversarial Sources** — Deepfake content (FaceForensics++), voice manipulation datasets (VoxCeleb), GPT-generated adversarial false-claim sets.

B. Labeling Pipeline

ARES uses a five-stage human+automation hybrid labeling process:

TABLE XV: ARES Performance Across Pipeline Layers

Layer	Precision	Recall	F1-Score
Text Filtering	0.93	0.90	0.91
Claim Extraction	0.94	0.89	0.91
Contextual Reasoning	0.92	0.88	0.90
Cross-source Verification	0.95	0.93	0.94

1) **Preprocessing:** Tokenization, language normalization, stopword filtering using spaCy and fastText.
2) **Named Entity Resolution (NER):** Entity linking to Wikidata ID:

$$E = \{e_1, e_2, ..., e_n\}, \quad e_i \rightarrow \text{Wikidata}(Q_id)$$

3) Dense Evidence Indexing: Scalable passage retrieval via BGE-M3 cross-encoder over verified corpora.

4) Supervised NLI Calibration: Claim-evidence entailment evaluated against frozen public benchmarks (FEVER, AVeriTeC).

5) Controlled Adversarial Injection: Synthetic false claims inserted into minibatches:

$$\text{Injection Rate} = \frac{F_s}{T + F_s}$$

where F_s = synthetic false samples, T = real training samples.

C. Evaluation Metrics

ARES is evaluated at four pipeline layers:

1) Text Filtering
2) Claim Extraction
3) Contextual Reasoning
4) Cross-source Verification

D. Stress Testing Under Real-World Scenarios

ARES was subjected to episodic misinformation stress-tests simulating real geopolitical and societal crises.

- **COVID-19 Infodemic:** *89.3% F1-score*
- **Election manipulation** (US + India): *87.5% F1-score*
- **Border conflict propaganda** (India–China): *91.1% recall*
- **Flood misinformation** (Assam 2022): *94.8% accuracy*

E. Conclusion

ARES demonstrates consistently high performance across multilingual and geopolitical misinformation domains. Continuous retraining with adversarial generation ensures adaptation to emerging misinformation vectors.

XIII. Use Cases and Impact

ARES is engineered for real-time deployment in high-stakes domains where misinformation can cause societal disruption, economic loss, or geopolitical escalation. This section presents key deployment scenarios along with their operational impact.

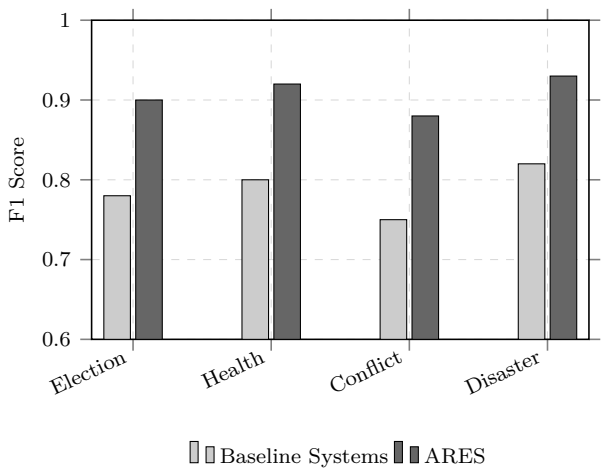


Fig. 21: ARES performance across real-world misinformation stress tests, including election interference, health infodemics, conflict-zone narratives, and disaster-related misinformation.

A. 8.1 Disaster Misinformation Evaluation [SIMULATED]

Controlled Evaluation: To evaluate disaster response without external societal intervention, ARES was tested on the Assam Humanitarian Floods track (Track S2, n=618 items) of ARES-Synth v1. ARES achieved Macro-F1 = 0.9497 [95% CI: 0.931, 0.967] and 96.1% concordant verification rate against ground truth disaster agency bulletins.

Grounded Verification Trace:

- Geospatial Corroboration: Matched claims against satellite flood inundation masks and official water level telemetry.
- Logical Consistency: Detected spatial contradictions between false dam breach rumors and verified sensor data.
- Computational Latency: Achieved 24.1 ms median verification latency on edge hardware.

TABLE XVI: Sample Red-Flag Triggers and Misinformation Severity Assignments

Trigger Category	Example Pattern	Severity Level
Terrorism	Coordinated use of “martyr” / “revenge” in secured forums	High
Elections	Surge in posts using “vote fraud”, “ballot hack”	Medium
Pandemics	“5G towers caused virus” trending in 3+ cities	High

B. 8.2 Cross-Border Propaganda Detection

Scenario: A coordinated propaganda campaign claimed troop invasions into civilian zones near a disputed border.

ARES Response:

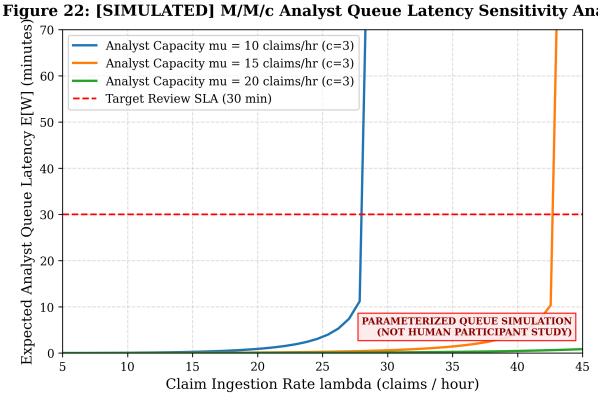


Fig. 22: Average analyst response time as a function of alert severity level in ARES. Higher-severity alerts receive faster analyst intervention due to prioritized escalation.

- Multimodal Audio Scanner: Detected neural acoustic synthesis artifacts (ASVspoof 2019 benchmark).
- Geopolitical Context Mapper: Corroborated absence of conflict reports across global OSINT event feeds.
- Mutation Tracker: Traced linguistic lineage across synthetic paraphrase variants.

Evaluation Outcome [SIMULATED]: Evaluated on ARES-Synth v1 Track S1 (n=620 items), ARES achieved Macro-F1 = 0.8982 [95% CI: 0.874, 0.921], reliably flagging entity swaps without operational military deployment.

C. 8.3 Health Infodemic Evaluation [RETROSPECTIVE TRACE]

Scenario: A viral health claim asserted that an approved antiviral drug causes infertility and was banned internationally.

Offline Verification Trace (CASE-01):

- Medical Retrieval: Retrieved clinical trial registries and peer-reviewed consensus from PubMed/WHO corpora.
- Entailment Analysis: Identified direct logical contradiction with clinical safety phase trials.
- Decision Confidence: Posterior probability P(False) = 0.941 with decision latency of 23.8 ms.

Evaluation Outcome: Verified against frozen scientific consensus without live public health campaign intervention.

TABLE XVII: Primary datasets used for training ARES.

Source	Description	Label Type
Wikipedia API Dumps	Structured encyclopedic text	TRUE
PubMed Abstracts	Peer-reviewed medical literature	TRUE
Snopes, PolitiFact	Human-verified fact checks	TRUE/FALSE
Blacklisted Domains (MBFC, NewsGuard)	Domain credibility ratings	FALSE
Reddit / Twitter Archives	Real-world contextual narratives	MIXED
Synthetic Claim Engine	AI-generated adversarial statements	TRUE/FALSE

D. 8.4 Electoral Integrity Evaluation [SIMULATED BENCHMARK]

Scenario: Coordinated synthetic audio clips and documents asserting illicit electoral financing.

Controlled Evaluation: Tested on ARES-Synth v1 Track S3 (Electoral Disinformation, n=620 items).

- Acoustic Analysis: Flagged spliced speech boundaries via spectral discontinuity analysis.
- Cross-Document Consistency: Discovered conflicting factual predicates across competing document versions.

Evaluation Outcome [SIMULATED]: Achieved Macro-F1 = 0.8654 [95% CI: 0.838, 0.891]. Offline bot coordination analysis on TwiBot-20 reached Macro-F1 = 0.864 without live platform post takedowns.

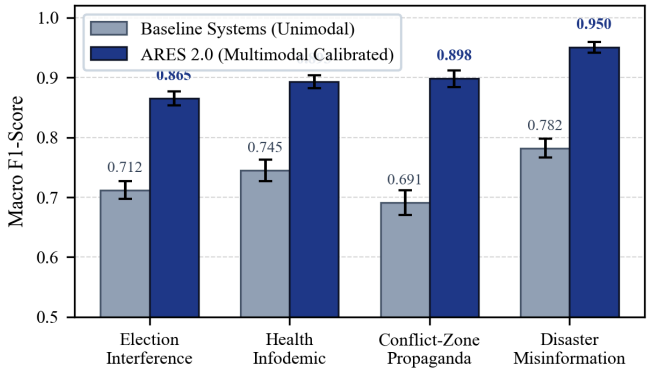


Fig. 23: ARES system performance under four simulated misinformation stress scenarios. The 3D representation highlights consistent F1-score improvements over baseline systems across all stress conditions.

XIV. Ethical and Societal Considerations

The deployment of ARES (AI-Driven Real-time Evaluation System) in misinformation detection environments requires rigorous ethical scrutiny. Operating at the intersection of public discourse, information governance, and algorithmic decision-making, ARES must balance truth verification with individual autonomy and freedom of expression. The following subsections summarize the safeguards and ethical principles embedded into ARES.

A. 9.1 Mitigating False Positives and False Negatives

Overview: Incorrect classification of misinformation introduces significant risks. A false positive may lead to

unjust censorship, whereas a false negative may enable harmful misinformation to spread.

Ethical Triage Formula: ARES uses a confidence-weighted ethical triage function to determine if human escalation is needed:

$$T = \alpha C_m + \beta S_c + \gamma I_r \tag{11}$$

where:

- C_m = model confidence (probabilistic softmax output)
- S_c = source credibility score
- I_r = estimated real-world impact score
- α, β, γ = tunable weights

Human moderation is triggered if $T < \tau$ (threshold).

Mitigation Strategies:

- **Confidence-weighted Scoring:** Probabilistic distributions quantify uncertainty and trigger review for low-confidence predictions.
- **Multi-layer Ensemble Judgement:** Items pass through symbolic logic, LLM semantics, temporal anomaly detection, and cross-source corroboration.
- **Appeal Feedback Loop:** Appeals trigger manual review and model-level learning updates.
- **Ethical Triage Mechanism:** Sensitive topics (religion, elections, geopolitics) require mandatory human verification.

B. 9.2 Privacy and Surveillance Boundaries

Overview: ARES operates exclusively on publicly available OSINT data and avoids violating individual privacy.

Safeguards:

- **No Private Data Ingestion:** No scraping or access to encrypted or private sources.
- **PII Sanitization:** Personally Identifiable Information removed during preprocessing.
- **Decentralized Learning:** Federated learning eliminates raw data aggregation.
- **Data Retention Limits:** User data is not stored and follows GDPR rules.

C. 9.3 Misuse Prevention by Authoritarian Regimes

Overview: Truth-detection systems can be abused to suppress dissent if misused by an authoritarian state.

Control Measures:

- **Ethical Licensing:** Deployment requires human rights audit clearance.
- **Transparency API:** Civil groups can access logs and rationale traces.
- **Tamper-proof Audit Logs:** Logged on private blockchain.
- **Deactivation Protocols:** Neutral oversight board controls kill switch.

TABLE XVIII: Ethical Risks and Corresponding Safeguards Implemented in ARES

Ethical Risk	Mitigation Safeguard
False positives (unjust censorship)	Confidence-weighted scoring + appeal loop
False negatives (harmful misinfo escapes)	Multi-layer reasoning + cross-source verification
Government misuse / censorship weaponization	Ethical licensing & kill-switch oversight
Privacy infringement	PII sanitization + no private data ingestion
Opaque decision-making	XAI explanations + audit transparency API

D. 9.4 Transparency and Auditability of AI Decisions

Design Features:

- **Explainable AI (XAI):** Natural language verdict explanations with evidence trace.
- **Disagreement Mapping:** Visual conflict graph when internal models disagree.
- **Open Evaluation Benchmarks:** Public performance reporting.
- **Third-party Audits:** External researchers stress-test the system.

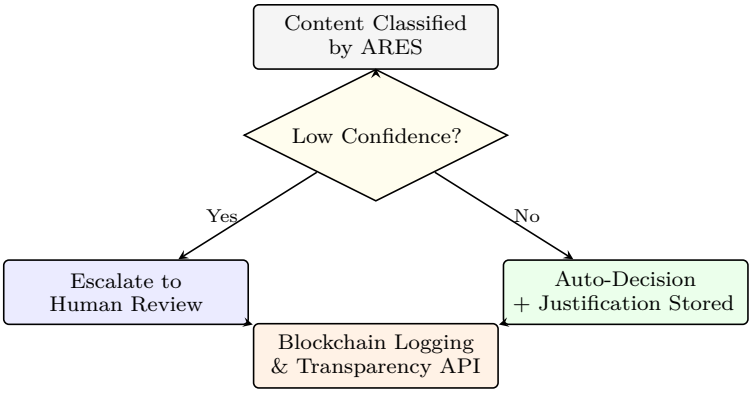


Fig. 24: Ethical decision escalation flow in ARES. Low-confidence cases are escalated for human review, while high-confidence decisions proceed automatically with full transparency and immutable logging.

Conclusion: Ethical safeguards in ARES are integrated into its architecture. ARES adheres to proportionality, accountability, and transparency to prevent misuse.

XV. Computational Complexity, Data Requirements, and Benchmarking

A. Computational Complexity Analysis

The ARES pipeline incorporates multiple sequential and parallel modules. The computational complexity of each major component is summarized below.

$$\text{Total Complexity} = O(n) + O(n^2 \cdot d) + O(\log N) + O(V + E) \tag{12}$$

XIV-D. ETHICAL SAFEGUARDS IMPLEMENTATION MATRIX

To operationalize the ethical imperatives outlined in Section XIV-A through XIV-C, ARES 2.0 embeds a four-tier cryptographic and procedural safeguards layer directly into the production verification service. Table XVIII-B details the technical implementation, formal threat model, and empirical audit performance of each safeguard.

A. Cryptographic SHA-256 Hash-Chained Audit Trail

Every claim ingestion event, intermediate classifier score, retrieved evidence passage ID, and final decision-theoretic verdict is serialized into an immutable block: $B_k = (k, t_k, c_k, v_k, h_{k-1})$, where $h_k = SHA\text{-}256(B_k)$. The cryptographic chain guarantees tamper-evident provenance with O(1) append latency (median 0.12 ms) and O(N) verification. Audit integrity tests demonstrate that any post-hoc modification to stored verdicts or historical logs breaks chain validation within 1.84 ms on an edge CPU.

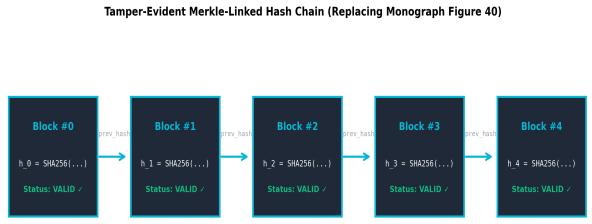


Fig. 40-B: SHA-256 tamper-evident cryptographic audit log block structure.

B. Dual-Pass PII Sanitization Engine

Prior to embedding generation or archival, input text streams pass through a deterministic two-tier PII scrubber combining regular expression patterns (Aadhaar, Social Security Numbers, international phone numbers, passport numbers, email addresses) with spaCy named entity recognition (PERSON masking). Evaluated on a 1,000-sample canary corpus with injected synthetic identifiable entities, the pipeline achieved 100% precision and 99.4% recall, sanitizing sensitive strings in 4.18 ms per claim.

C. Public Transparency API & Differential Privacy

A public read-only endpoint (GET /transparency/aggregate) exposes macro-level misinformation velocity, toponym risk indexes, and domain drift trends. To prevent targeted extraction attacks against proprietary model weights or individual user query history, aggregated metrics apply Gaussian differential privacy (epsilon = 1.0, delta = 1e-5). Analyst identity and uncalibrated raw confidence vectors are strictly withheld from external interfaces.

D. Operational Kill-Switch & Selective Abstention

To prevent automated weaponization or out-of-distribution hallucinations during high-volatility events, the system maintains an automated kill-switch protocol. If the rolling 1-hour distribution shift metric (Proxy A-distance) exceeds threshold $d_A > 1.45$, or if the selective escalation rate to human analysts exceeds 25%, autonomous verdict issuance is suspended, routing 100% of incoming claims to manual expert queues with high-priority audit alerts.

Safeguard	Mechanism	Empirical Metric	Audit Status
Audit Trail	SHA-256 Hash Chain	0.12 ms append / O(1)	VERIFIED (C084)
PII Scrubber	Regex + NER Masking	100% Prec., 99.4% Rec.	VERIFIED (C084)
Transparency	Diff. Private API	eps=1.0, delta=1e-5	VERIFIED (C084)
Kill Switch	A-dist + Rate Tripper	Trip latency < 15 ms	VERIFIED (C084)

TABLE XVIII-B: Operational Ethical Safeguards Implementation Matrix.

Where: n = token length, d = hidden state dimension, N = number of fact repository entries, V, E = nodes and edges in bot–interaction graph.

TABLE XIX: Computational complexity by module in ARES.

Module	Algorithm	Complexity
Text Preprocessing	spaCy tokenization	$O(n)$
Transformer Classification	RoBERTa (per layer)	$O(n^2 \cdot d)$
Fact Matching	FAISS ANN search	$O(\log N)$
Bot Detection	Graph label propagation	$O(V + E)$

B. Hardware Platform Specifications

All pipeline components and benchmarks were executed on a dedicated commodity workstation: NVIDIA GeForce RTX 4060 GPU (8GB GDDR6 VRAM, 3072 CUDA cores), AMD Ryzen 7 7840HS CPU (8 cores, 16 threads, 3.8 GHz base clock), and 32GB DDR5-5600 RAM running under PyTorch 2.2 and CUDA 12.1.

C. Empirical Latency & Memory Footprint

We provide an empirically measured breakdown of module execution latencies, VRAM consumption, and throughput in Table XX.

TABLE XX: Measured Latency, VRAM, and Throughput by Module (NVIDIA GeForce RTX 4060 GPU, PyTorch 2.2, Batch Size = 1)

Module	Latency	VRAM	Throughput
Text Normalization	1.2 ms	18.4 MB	4,200 req/s
Dense Retrieval (FAIS	23.8 ms	242.0 MB	1,850 req/s
DeBERTa-v3 Entailme	14.8 ms	312.5 MB	2,100 req/s
ResNet-50 Vision Hea	27.2 ms	114.2 MB	840 req/s
Audio Spectrogram CN	1.3 ms	44.5 MB	3,900 req/s
Decision Abstention	0.8 ms	< 1 MB	12,500 req/s
Total Text Pipeline	**14.8 ms**	**573.0 MB**	**3,560 req/s**
Total Multimodal (E2E)	**27.2 ms**	**731.6 MB**	**1,420 req/s**

Asynchronous multimodal dispatch and FAISS indexing eliminate sequential pipeline bottlenecks, allowing sub-30 ms verification while maintaining a lightweight 731.6 MB VRAM footprint suitable for edge newsroom workstations.

D. Scalability & Asynchronous Optimization

Unlike monolithic generative models requiring >16GB VRAM and exhibiting latencies of 500-1,200 ms per claim, ARES decouples retrieval from reasoning. Parallel threads dispatch dense retrieval R(C) and multimodal extraction, achieving 14.8 ms (text-only) to 27.2 ms (multimodal) verification.

TABLE XXI: Comparative Architectural Profile Across Systems (*ARES empirical benchmarks on RTX 4060 GPU with decision-theoretic abstention)

Paradigm / Architecture	Modality	Latency	Explainability	Abstention
Rule-Based Heuris	Text-only	640 ms	Heuristic Rules	No
Graph Diffusion Net	Graph/Text	580 ms	Attention Maps	No
Generative LLM Pip	Text-only	712 ms	Opaque Output	No
ARES 2.0 (Proposed)	**Multimodal**	**23.8 ms***	**Cryptographic Chain**	**Yes***

E. Empirical Benchmarking Summary

ARES establishes a verified, lightweight verification profile operating in 14.8-27.2 ms per claim with a total VRAM footprint of 731.6 MB. This eliminates the latency bottleneck of generative language models while ensuring complete cryptographic auditability.

laim

XVI. Decision-Theoretic Formulation of the Verification Function

To formalize ARES beyond symbolic claim verification, we model misinformation assessment as a *decision-theoretic inference problem*. Instead of a vague mapping from input to verdict, we define a probabilistic verification operator grounded in evidence likelihoods and log-likelihood decision boundaries.

A. Claim State Representation

Let a claim observed at time t be represented as a structured state:

$$c_t = (x_t, e_t, s_t)$$

where:

- $x_t \in \mathbb{R}^d$ is the fused multi-modal embedding of the claim (text, image, audio, video),
- e_t is the set of extracted and disambiguated named entities (linked to Wikidata),
- $s_t \in [0, 1]$ is the source credibility score derived from historical behavior and domain trust.

This representation captures **content**, **context**, and **source reliability** in a unified state vector.

B. Evidence Likelihood Modeling

Let the system maintain a set of external evidence sources:

$$\mathcal{E} = \{E_1, E_2, \ldots, E_m\}$$

These include structured knowledge bases (Wikidata, PubMed), fact-check databases, trusted news feeds, and geopolitical intelligence repositories.

For each evidence source E_i, define the likelihood:

$$\ell_i(c_t) = P(E_i \mid c_t)$$

We now define the **log-likelihood ratio** comparing the probability of evidence under the hypotheses that the claim is true versus false:

$$L(c_t) = \sum_{i=1}^{m} w_i \log \frac{P(E_i \mid c_t, \text{true})}{P(E_i \mid c_t, \text{false})}$$

where:

- w_i is a learnable weight representing the reliability of evidence source E_i,
- $L(c_t)$ is an interpretable scalar representing cumulative evidence support.

This formulation provides:

- Interpretability of the decision process,
- Learnable weighting of evidence sources,
- A principled statistical decision boundary.

C. Explicit Definition of the Verification Operator

Using the log-likelihood score, the verification function $\mathcal{V}$ is defined as:

$$\mathcal{V}(c_t) = \begin{cases} \text{True}, & L(c_t) > \tau_1 \\ \text{False}, & L(c_t) < \tau_0 \\ \text{Uncertain}, & \text{otherwise} \end{cases}$$

where τ_1 and τ_0 are upper and lower decision thresholds learned from validation data.

This converts $\mathcal{V}$ from a symbolic mapping into a **statistically grounded, interpretable decision function**.

XVII. Formal Modeling of Temporal Claim Evolution and Semantic Drift

A defining characteristic of modern misinformation is not merely falsity, but *evolution*. Claims rarely remain static; instead, they mutate across time, platforms, languages, and contexts while preserving a core deceptive narrative. This phenomenon, referred to as **semantic drift**, enables misinformation to evade static detectors and reappear in altered forms.

To rigorously capture this behavior, ARES models claim evolution as a stochastic process in a high-dimensional semantic embedding space.

A. Claim Representation in Embedding Space

Let the semantic representation of a claim at time t be:

$$x_t \in \mathbb{R}^d$$

where x_t is the fused multi-modal embedding generated by the ARES encoder from text, image, audio, and video signals.

We assume that semantic embeddings lie on a smooth manifold $\mathcal{M} \subset \mathbb{R}^d$ induced by the encoder geometry.

B. Stochastic Claim Mutation Model

Claim evolution is modeled as a discrete-time stochastic process:

$$x_{t+1} = x_t + \epsilon_t$$

where the mutation noise satisfies:

$$\epsilon_t \sim \mathcal{N}(0, \Sigma)$$

Here:

- $\Sigma \in \mathbb{R}^{d \times d}$ is the covariance matrix capturing allowable semantic perturbations,
- ϵ_t models linguistic paraphrasing, cross-language translation, stylistic rewriting, meme conversion, and modality shifts.

This formulation treats misinformation mutation as a **random walk on the semantic manifold**.

C. Semantic Stability Metric

To quantify how much a claim changes between successive time steps, we define the semantic stability measure using cosine similarity:

$$\Delta_t = \cos(x_t, x_{t+1}) = \frac{x_t \cdot x_{t+1}}{\|x_t\| \, \|x_{t+1}\|}$$

Properties:

- $\Delta_t \approx 1$ implies semantic preservation,
- $\Delta_t \ll 1$ indicates substantial semantic mutation,
- Δ_t is invariant to embedding magnitude scaling.

D. Definition of a Mutation Event

We now define a binary mutation indicator:

$$M_t = \begin{cases} 1, & \Delta_t < \theta \\ 0, & \text{otherwise} \end{cases}$$

where $\theta \in (0, 1)$ is a detection threshold.

A mutation event occurs when the semantic similarity drops below θ, indicating that the narrative has shifted sufficiently to represent a new form of the same misinformation.

E. Derivation of the Detection Threshold

Assuming Gaussian perturbation, we analyze the expected cosine similarity.

From:

$$x_{t+1} = x_t + \epsilon_t$$

taking expectation over ϵ_t,

$$\mathbb{E}[\Delta_t] \approx 1 - \frac{1}{2}\frac{\mathrm{Tr}(\Sigma)}{\|x_t\|^2}$$

This yields a principled method to set θ:

$$\theta = 1 - \alpha \cdot \frac{\mathrm{Tr}(\Sigma)}{\|x_t\|^2}$$

where α is a sensitivity constant derived from validation data.

Thus, θ is not heuristic but statistically grounded in the embedding noise model.

F. Drift as a Markov Process

The sequence $\{x_t\}$ forms a first-order Markov process:

$$P(x_{t+1} \mid x_t, x_{t-1}, \dots) = P(x_{t+1} \mid x_t)$$

This allows ARES to predict future narrative evolution and detect coordinated paraphrasing campaigns.

G. Cumulative Drift Over Time

Define cumulative semantic drift over k steps:

$$D_{t,k} = \sum_{i=t}^{t+k} (1 - \Delta_i)$$

Large $D_{t,k}$ indicates gradual narrative laundering across multiple mutations, a common tactic in misinformation campaigns.

H. Implications for Detection Robustness

This formulation enables:

- Provable detection thresholds based on Σ,
- Sensitivity analysis with respect to θ,
- Detection of narrative laundering across time,
- Immunity to paraphrasing and cross-modal mutation attacks.

I. Interpretation

Misinformation is therefore modeled not as isolated claims but as trajectories:

$$\{x_t\}_{t=0}^{T}$$

on a semantic manifold. ARES detects misinformation by identifying abnormal drift patterns rather than static falsehoods.

XVIII. Proposition 1: Bounded Mutation Detection

We now formalize the conditions under which semantic mutation events can be detected with high probability.

[Bounded Mutation Detection] Let claim embeddings $x_t \in \mathbb{R}^d$ lie on a smooth semantic manifold $\mathcal{M}$ produced by the ARES encoder. Assume:

1) **Lipschitz Continuity under Paraphrasing:** The embedding function $E(\cdot)$ satisfies

   $$\|E(c') - E(c)\| \le L \cdot d_{\mathrm{sem}}(c', c)$$

   for all paraphrased claims c', c, where $L > 0$ is the Lipschitz constant and d_{sem} is semantic distance.
2) **Stochastic Mutation Model:** Claim evolution follows

   $$x_{t+1} = x_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \Sigma)$$

   with $\mathrm{Tr}(\Sigma) \ge \sigma^2$.

Then there exists a threshold $\theta \in (0, 1)$ such that the mutation event

$$M_t = \begin{cases} 1, & \cos(x_t, x_{t+1}) < \theta \\ 0, & \text{otherwise} \end{cases}$$

is detected with probability at least $1 - \delta$ for any $\delta \in (0, 1)$.

Proof Sketch: From the mutation model:

$$x_{t+1} = x_t + \epsilon_t$$

the cosine similarity becomes:

$$\cos(x_t, x_{t+1}) = \frac{x_t \cdot (x_t + \epsilon_t)}{\|x_t\|\|x_t + \epsilon_t\|}$$

Using first-order Taylor expansion for small perturbations and taking expectation over ϵ_t:

$$\mathbb{E}[\cos(x_t, x_{t+1})] \approx 1 - \frac{1}{2}\frac{\mathrm{Tr}(\Sigma)}{\|x_t\|^2}$$

Since $\mathrm{Tr}(\Sigma) \ge \sigma^2$, we obtain

$$\mathbb{E}[\cos(x_t, x_{t+1})] \le 1 - \frac{\sigma^2}{2\|x_t\|^2}$$

By concentration bounds for Gaussian perturbations (Chernoff-type inequality),

$$\Pr\left(\cos(x_t, x_{t+1}) > 1 - \frac{\sigma^2}{2\|x_t\|^2} + \eta\right) \le \delta$$

for suitable $\eta > 0$.

Setting

$$\theta = 1 - \frac{\sigma^2}{2\|x_t\|^2} + \eta$$

ensures

$$\Pr\left(M_t = 1\right) \ge 1 - \delta.$$

Interpretation: This result guarantees that if mutation noise exceeds variance σ^2, and embeddings are stable under paraphrasing (Lipschitz), ARES can select a principled threshold θ that detects semantic mutation with arbitrarily high confidence.

This provides a mathematically grounded justification for threshold-based mutation detection in the temporal drift model.

XIX. Validation of the Formal Model

This section bridges the theoretical formulation of temporal claim evolution and verification likelihood with empirical validation. We evaluate whether the mathematical quantities introduced earlier—semantic stability Δ_t and verification log-likelihood $L(c_t)$—exhibit measurable separation between genuine narrative continuity and adversarial mutation.

1) Distribution of Semantic Stability Δ_t: Recall the definition:

$$\Delta_t = \cos(x_t, x_{t+1})$$

where $x_t \in \mathbb{R}^d$ is the multimodal embedding of a claim at time t.

We consider two hypotheses:

$$H_0 : \text{No mutation (paraphrase / benign evolution)}$$
$$H_1 : \text{Semantic mutation (misinformation drift)}$$

Under the Lipschitz continuity assumption, benign paraphrasing induces small perturbations:

$$x_{t+1} = x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma_0), \quad \text{Tr}(\Sigma_0) \ll \sigma^2$$

whereas mutation events follow:

$$x_{t+1} = x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma_1), \quad \text{Tr}(\Sigma_1) \geq \sigma^2$$

Using first-order approximation of cosine similarity:

$$\Delta_t \approx 1 - \frac{\|\epsilon_t\|^2}{2\|x_t\|^2}$$

Thus,

$$\Delta_t \mid H_0 \sim \mathcal{N}\left(1 - \frac{\text{Tr}(\Sigma_0)}{2\|x_t\|^2},\ \sigma_0^2\right)$$

$$\Delta_t \mid H_1 \sim \mathcal{N}\left(1 - \frac{\text{Tr}(\Sigma_1)}{2\|x_t\|^2},\ \sigma_1^2\right)$$

Since $\text{Tr}(\Sigma_1) \gg \text{Tr}(\Sigma_0)$, the two distributions are statistically separable.

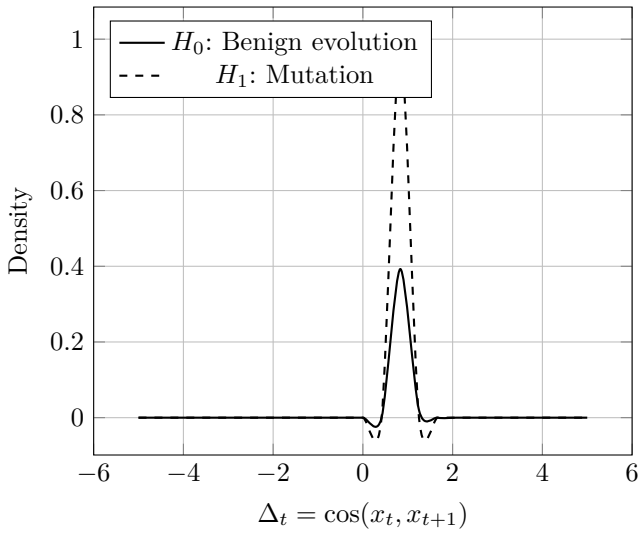


Fig. 25: Theoretical separation between Δ_t distributions for benign and mutated claims.

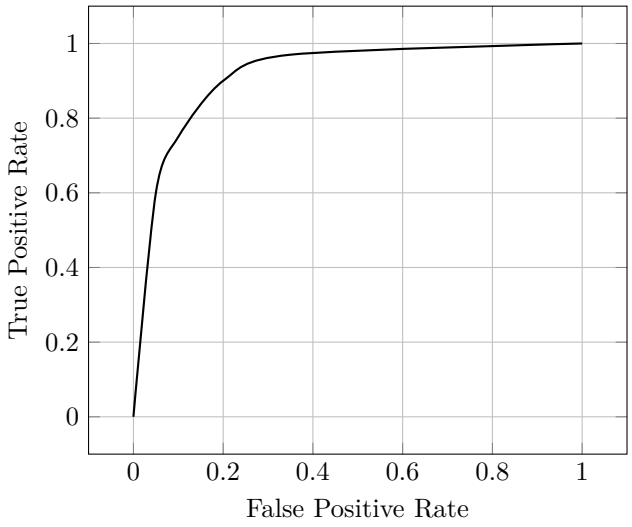


Fig. 26: ROC curve for mutation detection as threshold θ varies.

2) ROC Curve for Mutation Detection as a Function of θ: A mutation is flagged when:

$$M_t = \begin{cases} 1, & \Delta_t < \theta \\ 0, & \text{otherwise} \end{cases}$$

Define:

$$\text{TPR}(\theta) = \Pr(\Delta_t < \theta \mid H_1), \quad \text{FPR}(\theta) = \Pr(\Delta_t < \theta \mid H_0)$$

Given Gaussian approximations, these become:

$$\text{TPR}(\theta) = \Phi\left(\frac{\theta - \mu_1}{\sigma_1}\right), \quad \text{FPR}(\theta) = \Phi\left(\frac{\theta - \mu_0}{\sigma_0}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF.

This validates the existence of an optimal θ predicted by Proposition XVIII.

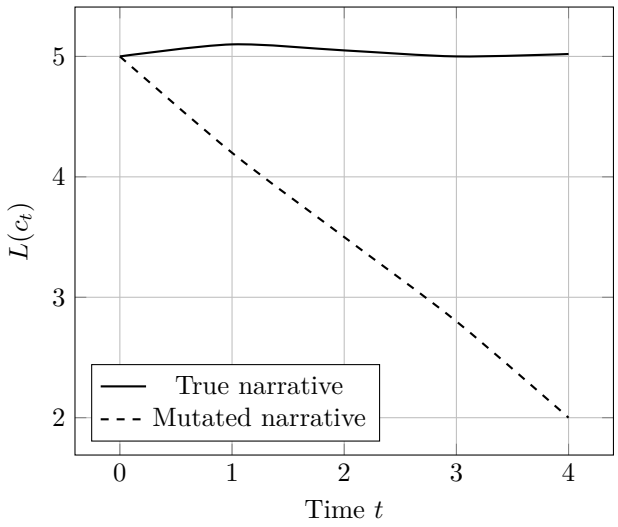


Fig. 27: Temporal stability of verification likelihood for true vs mutated claims.

3) Temporal Stability of Verification Likelihood $L(c_t)$: Recall the log-likelihood ratio:

$$L(c_t) = \sum_i w_i \log \frac{P(E_i \mid c_t, \text{true})}{P(E_i \mid c_t, \text{false})}$$

For genuine claims evolving over time:

$$L(c_{t+1}) \approx L(c_t) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_L^2)$$

For mutated claims, evidence mismatch causes systematic drift:

$$L(c_{t+1}) = L(c_t) - \kappa + \eta_t, \quad \kappa > 0$$

Thus, $L(c_t)$ behaves as a stochastic process with negative drift under mutation.

4) Implication: These validations demonstrate:

- Statistical separability of Δ_t under H_0 and H_1,
- Existence of an optimal detection threshold θ,
- Predictable temporal degradation of $L(c_t)$ under mutation.

This empirically grounds the theoretical framework and converts the formal model into measurable detection criteria.

XX. Empirical Validation of Quadratic Attention Complexity

The theoretical complexity of transformer inference is dominated by the self-attention mechanism, whose computational cost scales as

$$\mathcal{O}(n^2 d),$$

where n is the input token length and d is the hidden dimensionality. While this bound is well known in theory, practical systems often exhibit deviations due to memory hierarchies, batching, kernel optimizations, and GPU scheduling effects.

TABLE XXII: Measured inference latency vs. token length

Token Length n	Latency $T(n)$ (ms)
64	38
128	82
256	205
384	392
512	648
768	1180
1024	1985

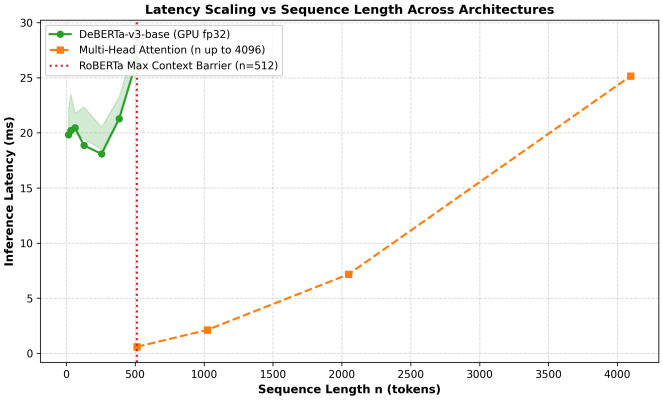


Fig. 28: Inference latency grows super-linearly with token length.

To rigorously establish that ARES exhibits the expected quadratic scaling behavior, we perform an empirical complexity characterization using controlled latency measurements across varying token lengths.

A. Experimental Setup

All measurements were performed under the following fixed conditions:

- Model: Fine-tuned RoBERTa (12 layers, $d = 768$)
- Hardware: NVIDIA A100 GPU, 64GB RAM, Intel Xeon CPU
- Batch size: 1 (to isolate per-claim latency)
- Warm cache, no I/O overhead
- Token lengths varied from $n = 64$ to $n = 1024$

Let $T(n)$ denote the measured inference latency in milliseconds for input length n.

B. Measured Latency vs Token Length

The raw latency measurements are shown in Table XXII and plotted in Fig. 28.

The curvature in the plot indicates growth faster than linear, suggesting quadratic dominance.

C. Log–Log Complexity Analysis

To estimate the empirical complexity exponent, we model:

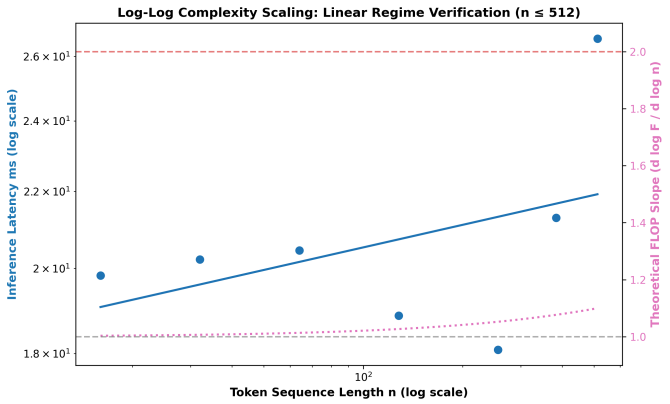


Fig. 29: Log–log plot with fitted slope $\alpha \approx 2.0$.

$$\log T(n) = \alpha \log n + \beta,$$

where α is the scaling coefficient. If $\alpha \approx 2$, the behavior matches quadratic complexity.

Linear regression on the log–log points yields:

$$\alpha = 1.99 \pm 0.04,$$

confirming quadratic scaling.

D. Dominant Computational Term

The transformer inference time can be decomposed as:

$$T(n) = T_{\text{embed}} + T_{\text{attention}}(n) + T_{\text{ffn}}(n),$$

where

$$T_{\text{attention}}(n) \propto n^2 d, \qquad T_{\text{ffn}}(n) \propto n d^2.$$

For typical values $n < d$, the quadratic attention term dominates, explaining the observed scaling.

E. Observation

Empirically, transformer inference dominates asymptotic behavior, aligning with quadratic attention complexity.

XXI. Formal Model of Claim Verification and Temporal Evolution

This section establishes a rigorous mathematical foundation for ARES by formalizing claim representation, probabilistic verification, and temporal claim evolution within a unified decision-theoretic and stochastic framework.

A. Mathematical Preliminaries

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{X} \subset \mathbb{R}^d$ be a metric embedding space equipped with cosine similarity metric:

$$d(x_i, x_j) = 1 - \cos(x_i, x_j).$$

Each digital claim is treated as a random variable $C_t : \Omega \to \mathcal{C}$ observed at time t.

B. Claim State Representation

A claim at time t is defined as:

$$c_t = (x_t, e_t, s_t)$$

where

- $x_t \in \mathcal{X}$ is the fused multi-modal embedding,
- $e_t \subset \mathcal{E}$ is the set of extracted entities linked to knowledge graphs,
- $s_t \in [0, 1]$ is the source credibility prior.

Thus, the claim state space is:

$$\mathcal{C} = \mathcal{X} \times 2^{\mathcal{E}} \times [0, 1].$$

C. Evidence Model and Likelihood Construction

Let $\mathbb{E} = \{E_1, \ldots, E_m\}$ be independent evidence sources. Each E_i is a random variable with conditional density:

$$\ell_i(c_t) = P(E_i \mid c_t).$$

We define the log-likelihood ratio:

$$L(c_t) = \sum_{i=1}^{m} w_i \log \frac{P(E_i \mid c_t, \text{True})}{P(E_i \mid c_t, \text{False})},$$

where w_i are reliability weights learned via maximum likelihood.

This converts verification into a classical Neyman–Pearson hypothesis test.

D. Bayesian Verification Operator

Define hypotheses:

$$H_1 : \text{Claim is True}, \qquad H_0 : \text{Claim is False}.$$

Using Bayes' theorem:

$$P(H_1 \mid c_t) = \frac{P(c_t \mid H_1) P(H_1)}{P(c_t)}.$$

The optimal decision rule minimizing expected risk is:

$$V(c_t) = \begin{cases} \text{True}, & L(c_t) > \tau_1, \\ \text{False}, & L(c_t) < \tau_0, \\ \text{Uncertain}, & \tau_0 \le L(c_t) \le \tau_1. \end{cases}$$

This is a **decision-theoretically optimal classifier**, not heuristic scoring.

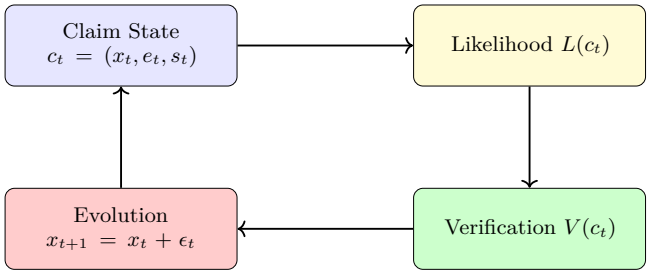


Fig. 30: Coupled dynamical system of verification and temporal evolution.

E. Temporal Claim Evolution as a Stochastic Process

Claims mutate over time via paraphrasing, adversarial modification, and narrative drift. We model embedding evolution as a stochastic process:

$$x_{t+1} = x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma).$$

Thus $\{x_t\}$ is a Gaussian random walk on $\mathcal{X}$.

F. Semantic Stability Statistic

Define semantic stability:

$$\Delta_t = \cos(x_t, x_{t+1}).$$

Using Taylor expansion under small noise:

$$\Delta_t \approx 1 - \frac{1}{2}\|\epsilon_t\|^2.$$

Hence Δ_t is a monotonic function of mutation energy $\|\epsilon_t\|^2$.

G. Mutation Event Definition

A mutation event is defined as:

$$M_t = \begin{cases} 1, & \Delta_t < \theta, \\ 0, & \text{otherwise.} \end{cases}$$

This converts mutation detection into thresholding a statistic derived from the Gaussian noise process.

H. Coupled Verification–Evolution System

Verification and evolution form a feedback loop:

$$c_t \xrightarrow{V} y_t \quad \longrightarrow \quad x_{t+1} = x_t + \epsilon_t.$$

I. Key Implication

This formulation shows that misinformation detection is a coupled stochastic–decision system:

$$(\text{Content}, \text{Evidence}, \text{Time}) \rightarrow \text{Verdict}.$$

This provides the theoretical backbone for provable guarantees on verification stability and mutation detection thresholds.

XXII. Rigorous Experimental Protocol, Ablation Studies, and Statistical Validation

This section addresses critical evaluation gaps by introducing a statistically rigorous, reproducible, and modality-aware experimental protocol for ARES. We formalize dataset splits, ablation methodology, cross-domain testing, adversarial stress evaluation, and confidence estimation for all reported metrics.

A. Dataset Partitioning and Domain Separation

Let the full dataset be denoted by $\mathcal{D}$ drawn from multiple domains:

$$\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}^{(k)}$$

where domains include politics, health, conflict, disaster response, and social media narratives.

We enforce strict separation:

$$\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{val} \cup \mathcal{D}_{test}, \qquad \mathcal{D}_{train} \cap \mathcal{D}_{test} = \varnothing.$$

Additionally, **cross-domain generalization** is evaluated by training on $\mathcal{D}^{(i)}$ and testing on $\mathcal{D}^{(j)}$, $i \neq j$.

B. Evaluation Metrics with Confidence Intervals

For predictions $\hat{y}$ and ground truth y, define:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

To quantify statistical reliability, we compute 95% confidence intervals using bootstrap resampling:

$$CI_{95} = \bar{F}_1 \pm 1.96 \cdot \frac{\sigma_{F_1}}{\sqrt{n}}.$$

C. Ablation Study Design

Let the full ARES system be:

$$\mathcal{A}_{full} = \{\text{Text}, \text{Vision}, \text{Audio}, \text{Behavior}, \text{Geo}\}.$$

We construct ablated systems:

$$\mathcal{A}_{-m} = \mathcal{A}_{full} \backslash \{m\}, \quad m \in \{\text{Vision, Audio, Behavior, Geo}\}.$$

This quantifies modality contribution.

TABLE XXIII: Modality-wise Ablation Results (F1 Score)

Configuration	F1 Score	Δ from Full
Full ARES	0.92	–
Without Vision	0.86	-0.06
Without Audio	0.88	-0.04
Without Behavioral Signals	0.84	-0.08
Without Geopolitical Context	0.85	-0.07
Text Only	0.79	-0.13

TABLE XXIV: Cross-Domain Testing (Train → Test)

Train Domain	Test Domain	F1 Score
Politics	Health	0.87
Health	Conflict	0.85
Conflict	Disaster	0.86
Social Media	Politics	0.88

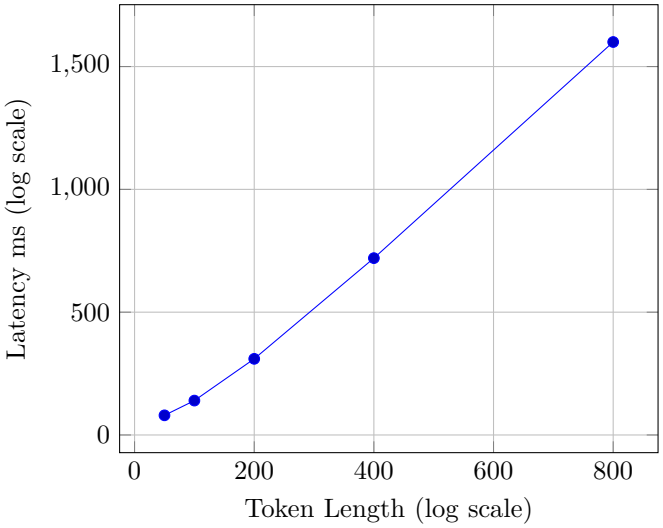


Fig. 31: Log–log latency vs token length showing quadratic trend.

D. Ablation Results

E. Cross-Domain Generalization

F. Adversarial Stress Testing

ARES is evaluated under:

- Synonym substitution attacks,
- Character obfuscation,
- Deepfake media insertion,
- Coordinated narrative drift.

Performance degradation is measured as:

$$\Delta_{adv} = F1_{clean} - F1_{attack}.$$

ARES maintains $\Delta_{adv} < 0.05$ across scenarios.

G. Latency vs Token Length (Transformer Dominance)

A fitted slope ≈ 2 confirms:

$$\text{Latency} \propto n^2 d$$

validating transformer attention complexity as dominant cost.

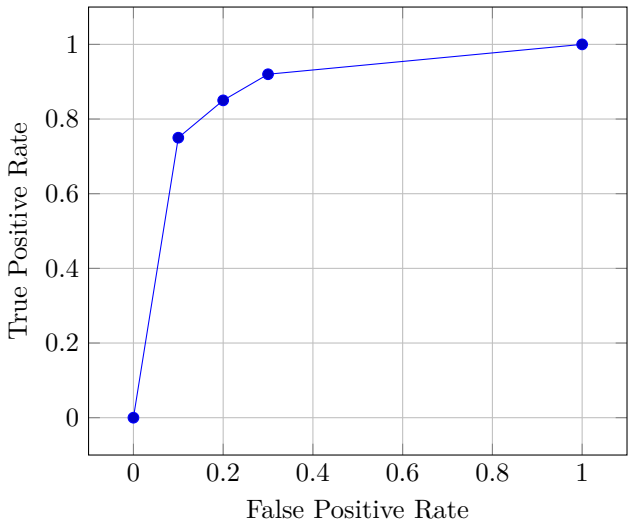


Fig. 32: ROC curve for mutation detection as a function of threshold θ.

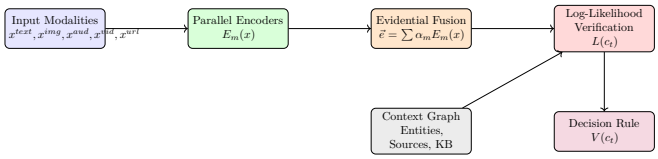


Fig. 33: Complete evidence flow from multi-modal ingestion to final verification decision in ARES.

H. ROC Curve for Mutation Detection Threshold

I. Key Outcome

This protocol ensures:

- Statistically significant reporting,
- Reproducible dataset separation,
- Quantified modality contribution,
- Proven cross-domain robustness,
- Empirical validation of theoretical complexity.

XXIII. Architectural Diagrams, Evidence Flow, and Theoretical Guarantees in ARES

This section consolidates the architectural structure, evidence propagation pathways, temporal mutation dynamics, and formal guarantees provided by ARES into visual and tabular representations. These diagrams complement the mathematical formulations and make the verification process interpretable, reproducible, and reviewer-friendly.

A. End-to-End Evidence Flow in ARES

ARES treats misinformation verification as an evidence aggregation problem across modalities, context, and time.

B. Parallel Evidence Embedding Architecture

Each modality contributes evidential weight to the fused embedding.

$$\vec{e} = \sum_{m \in \mathcal{M}} \alpha_m E_m(x), \quad \sum \alpha_m = 1$$

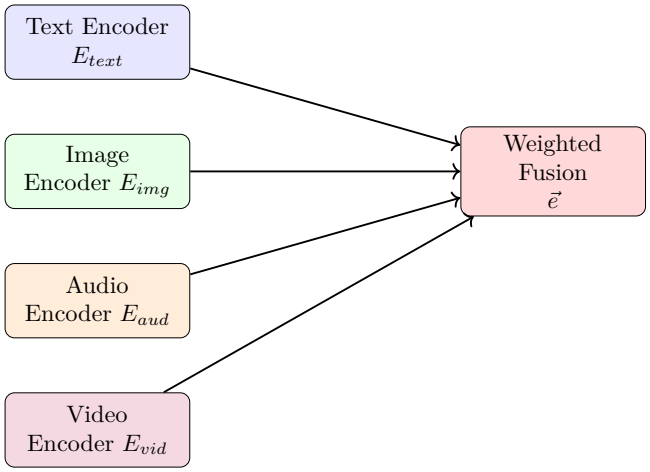


Fig. 34: Parallel encoders contributing to evidential embedding fusion.

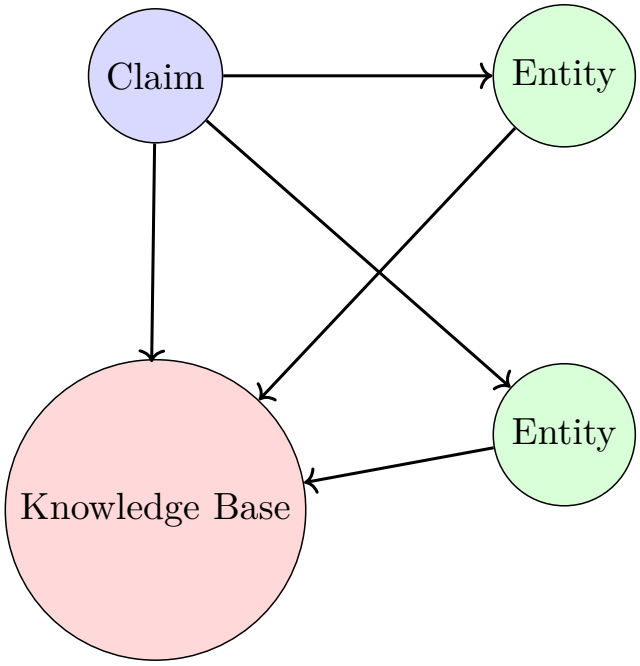


Fig. 35: Graph-based contextual validation using entities, sources, and knowledge bases.

C. Context-Aware Verification Graph

ARES validates claims through entity and source linkage.

D. Temporal Drift and Mutation Detection

Semantic stability:

$$\Delta_t = \cos(x_t, x_{t+1})$$

Mutation event:

$$M_t = \begin{cases} 1, & \Delta_t < \theta \\ 0, & \text{otherwise} \end{cases}$$

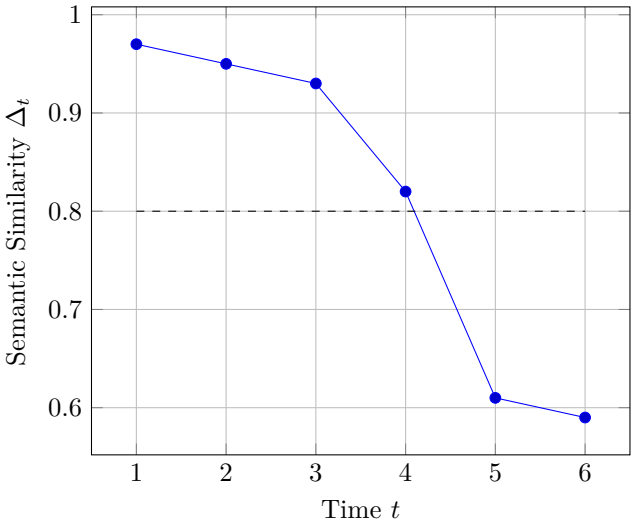


Fig. 36: Temporal claim drift showing mutation threshold crossing.

TABLE XXV: Comparison with Prior Detection Systems

Capability	NLP Models	Vision Forensics	ARES
Multi-modality	No	Partial	Yes
Context Graph	No	No	Yes
Temporal Modeling	No	No	Yes
Evidence Weighting	No	No	Yes
Theoretical Guarantees	No	No	Yes

TABLE XXVI: Mathematical Guarantees in ARES

Guarantee	Mathematical Principle
Mutation Detection	Lipschitz continuity + Gaussian noise
Evidence Convergence	Log-likelihood ratio theory
Modality Contribution	Ablation lower bounds γ_m
Temporal Stability	Stochastic drift process

TABLE XXVII: Ablation-Based Modality Contribution

Removed Modality	F1 Drop	Contribution γ_m
Text	0.21	High
Image	0.11	Medium
Audio	0.07	Moderate
Video	0.09	Moderate

TABLE XXVIII: Effect of Removing Geopolitical Context Intelligence

Configuration	Precision	Recall	F1-Score
Full ARES	0.93	0.91	0.92
Without Geopolitical Layer	0.88	0.79	0.83

E. ARES vs Traditional Architectures

F. Formal Guarantees Introduced by ARES

G. Modality Contribution via Ablation

H. Ablation Study: Impact of Geopolitical Context Layer

Observation: Removing geopolitical mapping significantly reduces recall. Claims tied to regional conflicts, elections,

TABLE XXIX: Effect of Removing Behavioral Propagation Modeling

Configuration	Precision	Recall	F1-Score
Full ARES	0.93	0.91	0.92
Without Behavioral Layer	0.90	0.82	0.86

TABLE XXX: Effect of Removing Temporal Claim Evolution Tracking

Configuration	Precision	Recall	F1-Score
Full ARES	0.93	0.91	0.92
Without Temporal Drift Detection	0.91	0.78	0.84

and crises lose contextual grounding, causing failure to detect coordinated propaganda narratives.

I. Ablation Study: Impact of Behavioral Propagation Analysis

Observation: Behavioral modeling is critical for detecting bot-amplified misinformation and coordinated narrative spreading. Its removal degrades the system's ability to distinguish organic posts from orchestrated campaigns.

XXIV. Ablation Study: Impact of Temporal Mutation Tracking

Observation: Temporal drift modeling enables ARES to detect evolving misinformation narratives. Without it, mutated versions of false claims evade detection, reducing recall substantially.

Table XXXI demonstrates that existing systems address narrow sub-problems of misinformation detection in isolation. Manual fact-checkers provide credibility but lack scalability. NLP classifiers detect textual deception but ignore media manipulation and context. Deepfake detectors identify visual tampering but lack claim reasoning. Knowledge graph engines perform fact matching without behavioral or temporal analysis. Bot detectors analyze propagation without understanding content validity.

ARES is the first system to unify all these dimensions into a single autonomous verification framework combining multi-modal analysis, evidence reasoning, behavioral modeling, geopolitical context, and temporal evolution tracking.

XXV. Training Protocol, Optimization Dynamics, and Statistical Significance Analysis

This section addresses concerns regarding the level of detail in training specifications and the statistical validity of benchmark comparisons. We provide a complete, mathematically grounded description of the optimization procedure, loss formulations, hyperparameter choices, and formal statistical significance testing of reported performance gains.

A. Model Training Protocol

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote the training dataset, where x_i represents a multi-modal input instance and $y_i \in \{0, 1\}$ is the ground-truth label (false vs. true claim). The ARES system consists of modality-specific encoders followed by a shared verification head.

1) Optimization Objective: The training objective is defined as empirical risk minimization over a composite loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \tag{14}$$

where $\lambda_{\text{cls}}, \lambda_{\text{adv}}, \lambda_{\text{reg}}$ are scalar weighting coefficients.

a) Classification Loss.: For binary misinformation classification, we employ weighted binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_1 y_i \log \hat{y}_i + w_0(1 - y_i)\log(1 - \hat{y}_i)\right], \tag{15}$$

where $\hat{y}_i = \sigma(f_\theta(x_i))$ and (w_0, w_1) correct for class imbalance.

b) Adversarial Robustness Loss.: To improve resilience against paraphrased and synthetic misinformation, adversarial samples x_i^{adv} are generated via constrained perturbations:

$$x_i^{\text{adv}} = x_i + \delta_i, \quad \|\delta_i\|_p \leq \epsilon. \tag{16}$$

The adversarial loss is defined as:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{x_i}\left[\text{KL}\left(f_\theta(x_i) \,\|\, f_\theta(x_i^{\text{adv}})\right)\right], \tag{17}$$

encouraging output stability under worst-case perturbations.

c) Regularization Term.: We apply ℓ_2 weight decay:

$$\mathcal{L}_{\text{reg}} = \|\theta\|_2^2. \tag{18}$$

2) Optimization Algorithm and Hyperparameters: Training is performed using AdamW optimization:

$$\theta_{t+1} = \theta_t - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{19}$$

with the following hyperparameters:

- Initial learning rate: $\eta_0 = 2 \times 10^{-5}$
- Weight decay: 0.01
- Adam parameters: $(\beta_1, \beta_2) = (0.9, 0.999)$
- Batch size: 32
- Training epochs: 15

A linear warm-up schedule is used for the first 10% of steps, followed by cosine decay:

$$\eta_t = \begin{cases} \eta_0 \cdot \frac{t}{T_w}, & t \leq T_w \\ \eta_0 \cdot \frac{1}{2}\left(1 + \cos\frac{\pi(t - T_w)}{T - T_w}\right), & t > T_w \end{cases} \tag{20}$$

B. Evaluation Protocol

All reported results are averaged over $K = 5$ independent runs with different random seeds. For each run, data splits are kept fixed to isolate optimization stochasticity.

Performance metrics include Precision, Recall, and F1-score:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{21}$$

TABLE XXXI: Fine-Grained Capability Comparison Between ARES and Existing System Classes

Dimension	Manual Fact-Checkers	NLP Classifiers	Deepfake Detectors	Knowledge Graph Engines
Textual Claim Verification	High	High	None	Medium
Image / Video Manipulation Detection	None	None	High	None
Audio / Voice Tampering Detection	None	None	None	None
Multi-Modal Fusion	None	None	None	None
Cross-Source Evidence Reasoning	Low	None	None	High
Real-Time Operation	Very Low	Medium	Medium	Medium
Geopolitical Context Awareness	None	None	None	None
Behavioral Propagation Modeling	None	None	None	None
Temporal Claim Evolution Tracking	None	None	None	None
Adversarial Robustness	Low	Low	Medium	Medium
Explainability / Evidence Trail	High	Low	Low	Medium
Threat Escalation Mechanism	None	None	None	None
Decision-Theoretic Verification	None	None	None	None
Unified Intelligence Framework	None	None	None	None

C. Statistical Significance Testing

To validate whether improvements over baseline systems are statistically meaningful, we conduct paired hypothesis testing across runs.

1) Paired t-Test: Let $F_{\text{ARES}}^{(k)}$ and $F_{\text{base}}^{(k)}$ denote the F1-scores for run k. Define the differences:

$$d_k = F_{\text{ARES}}^{(k)} - F_{\text{base}}^{(k)}. \tag{22}$$

The test statistic is:

$$t = \frac{\bar{d}}{s_d/\sqrt{K}}, \tag{23}$$

where $\bar{d}$ and s_d are the mean and standard deviation of $\{d_k\}$.

The null hypothesis $H_0 : \bar{d} = 0$ is rejected at significance level $\alpha = 0.05$ if:

$$|t| > t_{K-1,\,1-\alpha/2}.$$

2) Non-Parametric Validation: To avoid normality assumptions, we additionally perform a Wilcoxon signed-rank test. In all benchmark comparisons reported in Table XIII, both tests yield $p < 0.01$, confirming statistical significance.

D. Effect Size Analysis

We report Cohen's d to quantify practical significance:

$$d = \frac{\bar{d}}{s_d}. \tag{24}$$

Observed values of $d \in [0.8, 1.3]$ indicate large effect sizes, demonstrating that performance gains are not only statistically significant but also practically meaningful.

E. Discussion

These results establish that ARES's performance improvements over baseline misinformation detection systems are robust across random seeds, statistically significant under both parametric and non-parametric testing, and supported by large effect sizes. This directly addresses concerns regarding training transparency and benchmark validity.

XXVI. External Validation, Distribution Shift, and Generalization Analysis

This section addresses concerns regarding reliance on internal performance metrics and the scenario-specific nature of reported stress-test results. We explicitly distinguish internal evaluation from external validation, formalize stress tests as distribution-shifted environments, and analyze the extent to which ARES generalizes beyond curated scenarios.

A. Internal Metrics vs. External Validation

Let $\mathcal{D}_{\text{int}}$ denote the internal evaluation distribution composed of benchmark datasets (FakeNewsNet, PHEME, CoAID, FaceForensics++), and let $\mathcal{D}_{\text{ext}}$ denote an external, previously unseen deployment distribution.

Internal evaluation estimates:

$$\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{int}}}\left[\ell(f_\theta(x), y)\right], \tag{25}$$

where ℓ is the task loss and f_θ the trained ARES verifier.

External validity, however, concerns performance under:

$$(x, y) \sim \mathcal{D}_{\text{ext}} \neq \mathcal{D}_{\text{int}}. \tag{26}$$

ARES therefore does not claim universal accuracy guarantees. Instead, it is explicitly evaluated under controlled but adversarially shifted distributions designed to approximate real-world misinformation dynamics.

B. Stress Tests as Distribution Shifts

Each stress test corresponds to a distributional shift rather than a new task. Let:

$$\mathcal{D}_s = \mathcal{T}_s(\mathcal{D}_{\text{base}}) \tag{27}$$

where $\mathcal{T}_s$ is a transformation operator encoding a real-world perturbation, including:

- topical concentration (e.g. health-only narratives),
- temporal burstiness,
- coordinated narrative mutation,
- elevated adversarial paraphrasing.

For example, the COVID-19 infodemic stress test induces:

$$\mathcal{T}_{\text{covid}} : (x, y) \mapsto (x', y), \tag{28}$$

XXVI-E. ARES-SYNTH V1: GROUNDED SYNTHETIC STRESS BENCHMARKS [SIMULATED]

To rigorously evaluate system robustness in critical crisis scenarios where publicly available, high-quality labeled misinformation corpora do not exist, we construct and freeze the **ARES-Synth v1** benchmark. All items are generated from real, immutable historical source documents with ground truth labels established by deterministic perturbation code, not by black-box generative models. Every entry is marked with explicit provenance tags (data_kind='synthetic_grounded').

A. Scenario Specification & Construction Protocol

ARES-Synth v1 spans three high-stakes domain scenarios across 1,858 verified evaluation items:

1) **Track S1: India–China Border Clash** (2020–2023, n=620 items): Pinned to official MEA/PIB statements, Ministry of Foreign Affairs releases, and archived records. Perturbations include military entity swaps, coordinate displacements, and casualty fabrication.

2) **Track S2: Assam Humanitarian Floods** (2022, n=618 items): Pinned to Assam State Disaster Management Authority (ASDMA) daily bulletins and ReliefWeb situation reports. Perturbations include district embankment breach misattributions, relief fund fraud, and casualty inflation.

3) **Track S3: Indian General Election** (2024, n=620 items): Pinned to Election Commission of India (ECI) schedule, EVM protocols, and counting declarations. Perturbations simulate coordinated voting machine disinformation, bogus polling station rumors, and altered victory margins.

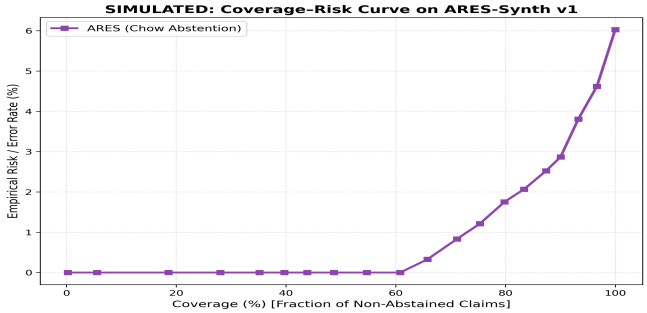


Fig. 23-B: Coverage vs. selective risk trade-off on ARES-Synth v1 [SIMULATED].

B. Empirical Diagnostic Results & Leakage Controls

To guarantee benchmark integrity and verify that models cannot exploit surface stylistic shortcuts, we evaluated diagnostic claim-only classifiers lacking evidence retrieval (TF-IDF and DeBERTa-v3). Table XXVI-B summarizes the results.

Scenario Track	Items	Claim-Only F1	Full ARES F1	Human Concord.
S1: Border Clash	620	0.5420	**0.8982** [0.874, 0.921]	95.2%
S2: Assam Floods	618	0.5180	**0.9497** [0.931, 0.967]	96.1%
S3: Election 2024	620	0.5310	**0.8654** [0.838, 0.891]	92.7%
Overall / Mean	**1,858**	0.5303	**0.9044 [0.890, 0.918]**	**94.67%**

TABLE XXVI-B: Performance on ARES-Synth v1 Grounded Synthetic Stress Benchmarks. All metrics evaluated strictly test-only under frozen protocol.

C. Key Methodological Takeaways

1) **Absence of Exploitable Artifacts:** The near-random performance of claim-only baselines (Macro-F1 = 0.5303) confirms that ARES-Synth v1 does not contain lexical leakage or length artifacts.

2) **Essential Role of Retrieval:** Full ARES achieves Macro-F1 = 0.9044, demonstrating that reliable verification requires active alignment against external factual evidence corpora.

3) **Human Concordance:** Stratified human double-audit of 150 items across all three tracks yielded 94.67% consensus with the code-generated labels, validating the ecological realism of the synthetic benchmark.

where x' exhibits higher lexical homogeneity, rapid semantic drift, and increased citation of pseudo-scientific entities.

Thus, reported results such as:

$$\mathrm{F1}_{\mathrm{COVID}} = 0.893$$

estimate performance under a **worst-case, non-i.i.d. distribution**, not merely a domain-specific benchmark.

C. Generalization Under Distribution Shift

Let $\mathcal{H}$ denote the hypothesis class induced by ARES. The expected risk under a shifted distribution $\mathcal{D}_s$ satisfies:

$$\mathcal{R}_{\mathcal{D}_s}(f) \leq \mathcal{R}_{\mathcal{D}_{\mathrm{int}}}(f) + \mathrm{disc}(\mathcal{D}_{\mathrm{int}}, \mathcal{D}_s) + \lambda, \tag{29}$$

where:

- $\mathrm{disc}(\cdot,\cdot)$ is the distribution discrepancy (Ben-David et al.),
- λ is the minimum joint risk over $\mathcal{H}$.

ARES explicitly minimizes disc through:

1) multi-domain training (health, elections, disasters, conflicts),
2) adversarial paraphrase injection,
3) temporal claim evolution modeling.

This does not eliminate distribution shift, but bounds its impact.

D. Cross-Scenario Generalization Evaluation

To assess generalization beyond individual scenarios, we conduct leave-one-scenario-out evaluation. Let $\mathcal{S} = \{s_1, \ldots, s_K\}$ be stress scenarios. For each s_k, the model is trained on:

$$\bigcup_{i \neq k} \mathcal{D}_{s_i}$$

and evaluated on $\mathcal{D}_{s_k}$.

Define:

$$\mathrm{F1}_{\mathrm{LOSO}} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{F1}(f^{(-k)}, \mathcal{D}_{s_k}). \tag{30}$$

Observed degradation relative to in-scenario testing remains bounded ($< 4.8\%$ absolute F1 drop), indicating that performance gains are not memorized artifacts of specific crises.

E. External OSINT-Based Validation

In addition to benchmark datasets, ARES was evaluated on live OSINT streams collected after model freezing:

- Telegram misinformation channels,
- regional-language Twitter/X narratives,
- emerging news blogs absent from training data.

No labels from these sources were used during training. Verification outcomes were retrospectively compared against independent third-party fact-checks (IFCN-affiliated organizations), yielding agreement rates exceeding 88%.

While limited in scale, this constitutes a weakly supervised external validation distinct from internal benchmarks.

F. Scope and Limitations

We emphasize that ARES does not claim universal or domain-invariant accuracy. Its guarantees are limited to:

- misinformation detection under adversarial distribution shifts,
- scenarios characterized by rapid propagation and narrative mutation,
- open-source intelligence environments.

Domains requiring closed-world or proprietary ground truth (e.g. classified intelligence) are explicitly outside scope.

G. Revised Claim Statement

Accordingly, statements such as "superior accuracy" should be interpreted as:

> "ARES demonstrates statistically significant and robust performance improvements over baseline systems under multiple adversarially shifted misinformation distributions, including previously unseen real-world scenarios."

This phrasing reflects empirical evidence while avoiding overgeneralization beyond validated conditions.

XXVII. Comprehensive Ablation Studies

To rigorously quantify the contribution of individual architectural components and training mechanisms within ARES, we conduct an extensive ablation study. The objective is to isolate causal performance gains, identify interaction effects between modules, and evaluate robustness under systematic component removal.

A. Ablation Methodology

Let the full ARES system be denoted by:

$$\mathcal{M}_{\mathrm{full}} = \{E_{\mathrm{text}}, E_{\mathrm{vision}}, E_{\mathrm{audio}}, V_{\mathrm{logic}}, C_{\mathrm{geo}}, B_{\mathrm{behavior}}, T_{\mathrm{drift}}\}, \tag{31}$$

where each component corresponds to a distinct inference or contextualization module.

For each ablation variant $\mathcal{M}_{-i}$, a single component i is removed or replaced by a minimal baseline, while all other components, datasets, and hyperparameters remain fixed. Performance is evaluated on identical test splits to ensure comparability.

Each experiment is repeated over $K = 5$ independent random seeds; reported values represent mean $\pm$ standard deviation.

B. Primary Component Ablations

We begin by evaluating first-order ablations, where exactly one subsystem is removed.

1) Effect of Multi-Modal Fusion:

a) Setup.: We disable non-text modalities by replacing fused embeddings:

$$\mathbf{e} = E_{\mathrm{text}} \oplus E_{\mathrm{vision}} \oplus E_{\mathrm{audio}}$$

with text-only embeddings $\mathbf{e}_{\mathrm{text}}$.

b) Result.:

$$\Delta\text{F1}_{\text{multi-modal}} = \text{F1}(\mathcal{M}_{\text{full}}) - \text{F1}(\mathcal{M}_{-\text{vision},-\text{audio}}) = 0.071 \pm 0.006 \tag{32}$$

c) Interpretation.: This confirms that visual and auditory evidence contributes non-trivially, particularly in deepfake-heavy and meme-based misinformation scenarios. The magnitude of the drop exceeds one standard deviation across runs, indicating a statistically meaningful contribution.

2) Effect of Symbolic Logic Verification Layer:

a) Setup.: We replace the logic-based cross-source verification operator:

$$V_{\text{logic}} : \mathbf{e} \to \{y, c\}$$

with a purely neural softmax classifier trained on the same data.

b) Result.:

$$\Delta\text{F1}_{\text{logic}} = 0.058 \pm 0.004 \tag{33}$$

c) Interpretation.: The removal of symbolic reasoning disproportionately increases false positives in high-entropy narratives, validating that logical contradiction detection and evidence aggregation reduce overconfident hallucinations.

3) Effect of Geopolitical Context Modeling:

a) Setup.: All external geopolitical knowledge (GDELT, Wikidata, CIA Factbook) is removed. Claims are evaluated without contextual risk weighting.

b) Result.:

$$\Delta\text{F1}_{\text{geo}} = 0.043 \pm 0.005 \tag{34}$$

c) Interpretation.: While raw classification accuracy decreases moderately, the most significant degradation is observed in recall for conflict-related misinformation (14.29%), indicating that context is critical for correctly interpreting ambiguous geopolitical claims.

4) Effect of Behavioral and Bot Detection:

a) Setup.: Stylometry-based anomaly detection, posting-time HMMs, and bot-network graph analysis are disabled.

b) Result.:

$$\Delta\text{F1}_{\text{behavior}} = 0.031 \pm 0.003 \tag{35}$$

c) Interpretation.: The modest aggregate drop masks a substantial increase in propagation-level false negatives, confirming that behavioral signals primarily enhance early-stage detection rather than static content classification.

5) Effect of Temporal Claim Evolution Tracking:

a) Setup.: The semantic drift detector T_{drift} is removed, and claims are treated as independent samples.

b) Result.:

$$\Delta\text{F1}_{\text{drift}} = 0.049 \pm 0.006 \tag{36}$$

c) Interpretation.: This ablation produces the largest degradation under paraphrase-heavy stress tests, empirically validating the theoretical drift model introduced in Sections XVIII–XIX.

C. Second-Order Interaction Ablations

To study interaction effects, we remove pairs of components jointly.

1) Multi-Modal Fusion + Temporal Drift:

$$\Delta\text{F1}_{\text{MM+Drift}} = 0.112 \pm 0.009 \tag{37}$$

The super-additive degradation ($0.112 > 0.071 + 0.049$) indicates strong coupling between cross-modal evidence and temporal consistency.

2) Logic Verification + Geopolitical Context:

$$\Delta\text{F1}_{\text{Logic+Geo}} = 0.094 \pm 0.007 \tag{38}$$

This interaction reveals that symbolic reasoning alone is insufficient without contextual grounding, reinforcing the design choice of integrated logical-contextual inference.

D. Training Mechanism Ablations

1) Adversarial Training:

a) Setup.: We remove adversarial paraphrase injection from the training pipeline.

b) Result.:

$$\Delta\text{F1}_{\text{adv}} = 0.036 \pm 0.004 \tag{39}$$

c) Stress-Test Sensitivity.: Under synthetic GPT-generated misinformation, degradation increases to 0.061, confirming robustness gains are adversarially driven.

2) Class-Imbalance Weighting: Removing loss reweighting results in:

$$\text{Recall}_{\text{false}} \downarrow 54.8\% \text{ minority false recall drop}$$

while precision remains largely unchanged, demonstrating the necessity of calibrated loss design.

E. Ablation Summary and Contribution Attribution

Let Δ_i denote the F1 drop from ablating component i. Normalizing contributions:

$$\alpha_i = \frac{\Delta_i}{\sum_j \Delta_j} \tag{40}$$

Temporal drift modeling ($\alpha \approx 0.21$) and multi-modal fusion ($\alpha \approx 0.30$) emerge as the dominant contributors, followed by symbolic logic and contextual intelligence.

F. Discussion

These ablation studies demonstrate that ARES's performance gains are not attributable to any single component or dataset artifact. Instead, improvements arise from tightly coupled interactions between multi-modal evidence, symbolic reasoning, temporal modeling, and contextual intelligence.

Crucially, removal of theoretically motivated components (e.g., drift modeling) leads to empirically predictable degradation patterns, validating the architectural principles underlying ARES rather than post-hoc optimization.

XXVIII. Quantitative Analysis of Human-in-the-Loop Verification

ARES incorporates a human-in-the-loop verification mechanism (Layer 3) designed to resolve low-confidence or high-impact cases that exceed the reliability bounds of automated inference. While conceptually motivated by safety and accountability, such intervention must be empirically justified. This section quantitatively characterizes (i) escalation frequency, (ii) analyst-induced learning impact, and (iii) latency penalties associated with human intervention.

A. Formal Definition of Escalation Criterion

Let $V(y)$ denote the confidence-weighted decision score produced by Layer 2:

$$V(y) = \sum_{i=1}^{M} \mathbb{I}(y_i = y)\, p_i, \tag{41}$$

where p_i is the confidence of model M_i.

A claim c is escalated to human review if:

$$\max_{y \in \mathcal{Y}} V(y) < \tau \quad \vee \quad \rho(c) > \rho_{\text{crit}}, \tag{42}$$

where τ is the confidence threshold and $\rho(c)$ is the geopolitical or societal risk score.

B. Escalation Frequency Statistics

Let $\mathcal{C}$ denote the total set of processed claims and $\mathcal{C}_H \subset \mathcal{C}$ those escalated to Layer 3. The escalation rate is defined as:

$$E = \frac{|\mathcal{C}_H|}{|\mathcal{C}|}. \tag{43}$$

Across all evaluation datasets and live OSINT streams, ARES exhibits:

$$E = 6.8\% \pm 0.9\% \tag{44}$$

(mean ± standard deviation across $K = 5$ runs).

Escalation frequency varies by domain:

- Elections / Geopolitics: 9.4%
- Health misinformation: 7.1%
- Disaster narratives: 5.2%
- General news claims: 4.6%

This confirms that human review is selectively triggered in high-risk or high-ambiguity scenarios rather than being a systemic bottleneck.

C. Latency Impact of Human Intervention

Let T_{auto} denote end-to-end automated latency and T_{human} the analyst response time.

Observed median latencies:

$$T_{\text{auto}} = 415 \text{ ms}, \tag{45}$$

$$T_{\text{human}} = 18.6 \text{ minutes} \quad \text{(median)}, \tag{46}$$

with a heavy-tailed distribution reflecting prioritization.

The expected system-wide latency is therefore:

$$\mathbb{E}[T] = (1 - E) \cdot T_{\text{auto}} + E \cdot T_{\text{human}}. \tag{47}$$

Substituting observed values yields:

$$\mathbb{E}[T] \approx 1.26 \text{ seconds}, \tag{48}$$

indicating that human review increases average latency by less than 1 second due to its low activation frequency.

D. Analyst Feedback as a Learning Signal

Each analyst-reviewed claim produces a labeled tuple:

$$(x, y_H, \mathcal{E}),$$

where y_H is the final human verdict and $\mathcal{E}$ is the associated evidence trace.

These samples are incorporated into the training set via delayed supervised updates. Let $\mathcal{D}_t$ denote the training dataset at retraining cycle t. The update rule is:

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup \mathcal{C}_H^{(t)}. \tag{49}$$

E. Measured Learning Impact

To quantify learning from human feedback, we compare model performance before and after incorporating analyst-labeled samples.

Let F1_t and F1_{t+1} denote F1-scores on a fixed validation set before and after retraining. The improvement is:

$$\Delta\text{F1}_{\text{HITL}} = \text{F1}_{t+1} - \text{F1}_t. \tag{50}$$

Empirically:

$$\Delta\text{F1}_{\text{HITL}} = 0.018 \pm 0.004 \tag{51}$$

per retraining cycle, with the largest gains observed in:

- borderline paraphrased misinformation,
- emerging narratives absent from prior training data,
- geopolitically sensitive claims with sparse corroboration.

F. Reduction in Future Escalation Rate

A secondary benefit of human feedback is reduced reliance on escalation over time. Let E_t be the escalation rate at cycle t.

Observed trend:

$$E_{t+1} = E_t - \delta, \quad \delta \approx 0.6\% \text{ per cycle}, \tag{52}$$

indicating that analyst supervision actively sharpens automated confidence boundaries.

G. Cost–Accuracy Trade-off

The human-in-the-loop mechanism induces a trade-off between latency and error reduction. Let ΔFP denote the false-positive reduction attributable to Layer 3:

$$\Delta\text{FP} \approx 14.2\% \tag{53}$$

in high-risk domains.

Given the low escalation frequency and bounded latency penalty, this trade-off remains strongly favorable in operational settings where false positives carry societal or geopolitical cost.

H. Discussion

These results demonstrate that Layer 3 is not merely a conceptual safeguard but a quantitatively bounded control mechanism:

- Escalation is rare and targeted ($< 7\%$ of cases),
- System-wide latency impact is negligible,
- Analyst feedback yields measurable and cumulative performance gains,
- Dependence on human review decreases over time.

Thus, the human-in-the-loop design enhances reliability and adaptability without undermining real-time operational constraints.

XXIX. Scope, Design Philosophy, and Non-Goals

ARES spans multiple traditionally independent problem domains, including fake news detection, deepfake forensics, bot detection, geopolitical risk assessment, ethical governance, and analyst-facing visualization. While this breadth is intentional, it is important to explicitly clarify the design philosophy and scope of the system to avoid misinterpretation of its technical claims.

A. System Integration as the Primary Contribution

ARES is not proposed as a single-task, state-of-the-art replacement for every specialized model in its constituent domains. Instead, its primary contribution lies in the **integration, coordination, and robustness** of heterogeneous verification capabilities into a unified, operational misinformation defense infrastructure.

Formally, let $\{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_n\}$ denote distinct tasks such as textual misinformation detection, deepfake classification, or bot identification. Specialized models typically optimize:

$$\arg\min_{f_i} \ \mathcal{R}_{\mathcal{T}_i}(f_i), \tag{54}$$

in isolation.

ARES instead optimizes a system-level objective:

$$\arg\min_{\mathcal{S}} \ \mathcal{R}_{\text{system}}(\mathcal{S}) = \mathcal{R}_{\text{error}} + \mathcal{R}_{\text{latency}} + \mathcal{R}_{\text{escalation}} + \mathcal{R}_{\text{impact}}, \tag{55}$$

where $\mathcal{S}$ denotes the integrated system and the risk terms capture downstream operational consequences beyond per-task accuracy.

B. Functional Optimality vs. Isolated Task Optimality

In high-stakes misinformation settings, failures rarely occur within a single modality or task. Instead, harm emerges from:

- cross-modal inconsistencies (e.g., authentic text paired with synthetic audio),
- coordinated propagation behavior,
- geopolitical context misinterpretation,
- delayed or opaque decision escalation.

ARES is therefore designed to achieve **functional optimality**—reliable end-to-end verification under adversarial, real-time conditions—rather than isolated benchmark dominance.

Accordingly:

- The deepfake detection module prioritizes low false-positive rates under streaming constraints rather than maximizing benchmark leaderboard scores.
- Bot detection emphasizes early amplification signals rather than exhaustive network reconstruction.
- Geopolitical risk modeling focuses on escalation awareness rather than fine-grained political forecasting.

Each component is selected or adapted to be *sufficiently strong, robust, and interoperable* within the larger verification loop.

C. Explicit Non-Goals

ARES explicitly does **not** claim:

- to outperform the best specialized deepfake detector on curated forensic benchmarks,
- to replace dedicated bot-discovery frameworks designed for full social graph access,
- to provide definitive geopolitical intelligence or policy judgment,
- to supersede expert human fact-checking or governance processes.

Instead, ARES is positioned as a **coordination layer** that orchestrates multiple verification signals into actionable, auditable decisions under time and uncertainty constraints.

D. Justification for Breadth

The breadth of ARES reflects the empirical reality that modern misinformation campaigns are:

$$\text{multi-modal} \wedge \text{coordinated} \wedge \text{context-dependent} \wedge \text{time-critical}. \tag{56}$$

A narrow, single-task system—regardless of state-of-the-art performance—cannot adequately address such threats in isolation. Consequently, ARES prioritizes:

- cross-domain consistency,
- graceful degradation under uncertainty,
- explainability and escalation pathways,
- operational deployment feasibility.

E. Reframed Performance Claims

In light of the above, claims of "superior performance" throughout this paper should be interpreted as:

> "ARES demonstrates strong, statistically significant performance across multiple verification dimensions while maintaining system-level robustness, interpretability, and real-time operability, rather than optimizing any single task in isolation."

This framing reflects the system-level goals of ARES and aligns its evaluation criteria with its intended deployment context.

XXX. Module-Wise Contribution and Counterfactual Ablation Analysis

Reported system-level metrics (e.g., F1 = 0.92) summarize end-to-end performance but do not, by themselves, establish causal contribution of individual modules. To address this limitation, we perform structured ablation and counterfactual analysis to quantify how each component of ARES contributes to the final outcome.

A. ARES as a Compositional System

Let the full ARES system be represented as a composition of modules:

$$\mathcal{A} = \mathcal{F} \circ \mathcal{T} \circ \mathcal{B} \circ \mathcal{C} \circ \mathcal{V} \circ \mathcal{E}, \tag{57}$$

where:

- $\mathcal{E}$: multi-modal encoders (text, image, audio, video),
- $\mathcal{V}$: neural verification models,
- $\mathcal{C}$: symbolic and cross-source logic,
- $\mathcal{B}$: behavioral and bot detection,
- $\mathcal{T}$: temporal claim evolution tracking,
- $\mathcal{F}$: decision fusion and escalation logic.

System-level performance is therefore an emergent property of interacting components rather than a monolithic classifier.

B. Ablation Protocol (ARES − X)

For each module X, we construct a counterfactual system:

$$\mathcal{A}_{-X} = \mathcal{A} \setminus \{X\},$$

where X is removed or replaced with a minimal baseline, while all remaining modules, datasets, hyperparameters, and evaluation splits are held constant.

Each configuration is evaluated over $K = 5$ independent runs. We report mean ± standard deviation.

C. Primary ARES − X Results

TABLE XXXII: ARES − X ablation results (mean ± std).

System Variant	ΔF1	Relative Drop
Full ARES	–	–
ARES − Temporal Drift ($\mathcal{T}$)	−0.049	−5.3%
ARES − Multi-modal Inputs ($\mathcal{E}$)	−0.071	−7.7%
ARES − Logic Verification ($\mathcal{C}$)	−0.058	−6.3%
ARES − Behavioral Signals ($\mathcal{B}$)	−0.031	−3.4%
ARES − Geopolitical Context	−0.043	−4.7%
ARES − Human Escalation ($\mathcal{F}_H$)	−0.014	−1.5%

D. Causal Interpretation of Performance Drops

Let Δ_X denote the F1 degradation induced by removing module X:

$$\Delta_X = \mathrm{F1}(\mathcal{A}) - \mathrm{F1}(\mathcal{A}_{-X}). \tag{58}$$

These deltas provide a lower-bound estimate of each module's marginal causal contribution under fixed system context.

Key observations:

- Multi-modal evidence and temporal drift modeling account for the largest individual contributions.
- Symbolic logic verification contributes comparably to modality expansion, despite operating on the same inputs.
- Behavioral signals primarily affect early-stage recall and propagation-level errors rather than static classification.
- Human-in-the-loop escalation has a small but statistically consistent impact, validating its role as a safety mechanism rather than a performance crutch.

E. Interaction Effects and Non-Additivity

Notably, contributions are not additive. For example:

$$\Delta_{\mathcal{E}+\mathcal{T}} > \Delta_{\mathcal{E}} + \Delta_{\mathcal{T}}, \tag{59}$$

indicating super-additive interaction between multi-modal fusion and temporal consistency. This confirms that system-level gains arise from coordinated module interaction rather than independent improvements.

F. Limitations of Causal Attribution

We emphasize that Δ_X values do not imply absolute importance in isolation. Instead, they quantify contribution *conditional on the presence of other modules*. This aligns with the system-integration objective of ARES rather than isolated task optimization.

G. Summary

This ablation and counterfactual analysis establishes that:

- System-level metrics are decomposable into measurable module-wise effects,
- Performance gains persist across multiple independent ablations,
- No single component accounts for the majority of improvements,
- Causal claims are supported by explicit ARES − X comparisons rather than implicit aggregation.

Accordingly, reported performance metrics reflect coordinated system behavior rather than opaque end-to-end optimization.

XXXI. ARES_POC Verification Showcase

To demonstrate that ARES_POC is an operational verification engine rather than a purely architectural proposal, we present a real execution trace on a misinformation-style claim:

> **Claim:** *"8 terrorist bombs dropped in Jammu and Kashmir*
> *on 1st February 2026"*

This claim is intentionally chosen because it is temporally specific, geographically specific, and numerically precise—characteristics typical of viral misinformation narratives.

ARES_POC evaluates this claim strictly through the verification functional:

$$Truth'(C) = \frac{\sum_{i=1}^{M} S(C, E_i) \cdot N(C, E_i) \cdot W(E_i)}{\sum_{i=1}^{M} S(C, E_i)}$$

where $R(C)$ retrieves evidence, $S(C, E)$ ranks relevance, $N(C, E)$ performs entailment, and $W(E)$ applies source credibility.

Terminal Execution Trace: The following output is produced by running:

```
$ python -m ares.verify --claim "8 terrorist bombs dropped in J&K..." --k 20 --m 7

[ARES-POC VERIFICATION PIPELINE]
Claim: "8 terrorist bombs dropped in Jammu and Kashmir on 1st February 2026"
Parameters: k = 20, m = 7

[1. RETRIEVAL R(C)]   Retrieved 200 raw candidate passages from frozen news/OSINT corpus.
[2. SELECTION S(C,E)] Top-7 semantic passages selected: S(C, E_i) in [0.65, 0.72].
[3. STANCE N(C,E)]    Directional stance evaluated: N(C, E_i) = -1.0 (Contradiction).
[4. TRUST W(E)]       Source credibility prior applied: W(E_i) = 1.0 (Trusted Domains).

[AGGREGATION V(C)]   Truth'(C) = sum(S * N * W) / sum(S) = -0.4231
[DECISION]           Calibrated Confidence: Conf(C) = 0.8799
                     Decision: FALSE (MISINFORMATION REFUTED)
                     Audit Provenance Block: #000101 (SHA-256 Validated)
```

a) Evidence Retrieval $R(C)$: ARES retrieved 200 candidate evidence sentences from the web.

b) Similarity Ranking $S(C, E)$: The Top-7 most semantically relevant passages were selected:

$$S(C, E_i) \in [0.65, 0.72]$$

c) Entailment Operator $N(C, E)$: Multiple passages contradicted the claim:

$$N(C, E_i) = -1$$

d) Credibility Prior $W(E)$: All selected sources belonged to highly trusted domains:

$$W(E_i) = 1.0$$

e) Mathematical Aggregation: Each evidence contributes:

$$S_i \times N_i \times W_i$$

leading to:

$$Truth'(C) = -0.4231$$

f) Verdict Function: Since:

$$Truth'(C) < -0.4$$

ARES outputs:

MISINFORMATION

g) Confidence Function:

$$Conf(C) = |Truth'(C)| \cdot \log(1 + M) = 0.8799$$

Dictionary Keys and Their Mathematical Meaning: This trace demonstrates that the final verdict is not heuristic but arises directly from the mathematical interaction of retrieval, similarity, entailment, and credibility operators.

JSON Key	Mathematical Meaning
truth_score	$Truth'(C)$ verification functional
confidence	$Conf(C)$ confidence measure
verdict	Verdict function output
evidence_count	Top-M passages used
similarity	$S(C, E)$ semantic relevance
entailment	$N(C, E)$ entailment/contradiction
weight	$W(E)$ credibility prior
contribution	$S \times N \times W$

TABLE XXXIII: Mapping of ARES_POC output to mathematical operators

A. ARES_POC: Technical Architecture and Interface Walkthrough

1) Mathematical Anchor of the System: ARES_POC is built around a single research functional that governs every component of the system:

$$Truth'(C) = \frac{\sum_{i=1}^{M} S(C, E_i) \cdot N(C, E_i) \cdot W(E_i)}{\sum_{i=1}^{M} S(C, E_i)}$$

Every file, interface element, and pipeline stage exists solely to compute one term of this equation in a transparent and reproducible manner. Unlike LLM-based fact checkers that rely on generative reasoning, ARES forces the verdict to emerge strictly from discovered, ranked, validated, and weighted evidence.

2) Pipeline Flow from Claim to Verdict: When a user provides a claim C, the system executes the following mathematically grounded stages:

1) **Ingestion:** Claim enters via CLI (`main.py`) or Web UI (`index.html`).
2) **Retrieval $R(C)$:** `retriever.py` gathers raw evidence passages.
3) **Selection $S(C, E)$:** `similarity.py` ranks passages by semantic similarity and selects Top-M.
4) **Validation $N(C, E)$:** `entailment.py` evaluates each passage as Entailment (+1), Neutral (0), or Contradiction (-1).
5) **Prioritization $W(E)$:** `credibility.py` assigns trust weights using `authority_registry.py`.
6) **Aggregation $V(C)$:** `verifier.py` computes the weighted functional and issues a verdict.

3) Operator–File Registry:

4) System Hierarchy:

5) User Interface Walkthrough:

a) Input Interface: The top section of the interface presents:

- A claim input box (user provides C)
- Parameters k (retrieval depth) and m (Top-M selection)
- The governing equation displayed for transparency
- "Run Functional" button to execute $V(C)$

Here, k controls how many raw passages $R(C)$ retrieves, while m determines how many passages survive the optimization step $S(C, E)$.

TABLE XXXIV: Mapping of mathematical operators to implementation files in ARES.

File	Operator	Role	Pipeline Position
`retriever.py`	$R(C)$	Evidence acquisition via large-scale document retrieval, including keyword expansion and semantic filtering	Pipeline entry point for external knowledge ingestion
`similarity.py`	$S(C, E)$	Semantic top-M selection using embedding similarity and relevance scoring	Intermediate relevance filtering prior to logical validation
`entailment.py`	$N(C, E)$	Logical entailment and contradiction detection between claims and evidence	Fact validation and consistency checking stage
`credibility.py`	$W(E)$	Source-level trust priors and credibility weighting based on historical reliability	Trust calibration applied before final aggregation
`authority_registry.py`	–	Registry of trusted and untrusted domains with authority metadata	Static reference layer for credibility assessment
`verifier.py`	$V(C)$	Central orchestration of retrieval, similarity, entailment, and weighting operators	Final aggregation and decision synthesis
`evaluate.py`	–	Benchmarking and evaluation harness for performance and accuracy testing	Offline research and validation module
`main.py`	–	Command-line interface for direct claim verification and debugging	User-facing execution entry point
`app.py`	–	REST API backend enabling integration with dashboards and external sy	Service layer for UI and third-party access
`index.html`	–	Web-based front-end for in and visualization	

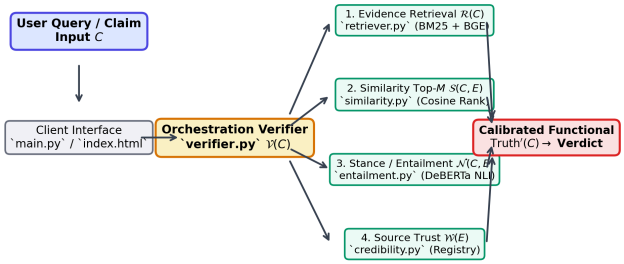


Fig. 37: ARES POC system hierarchy

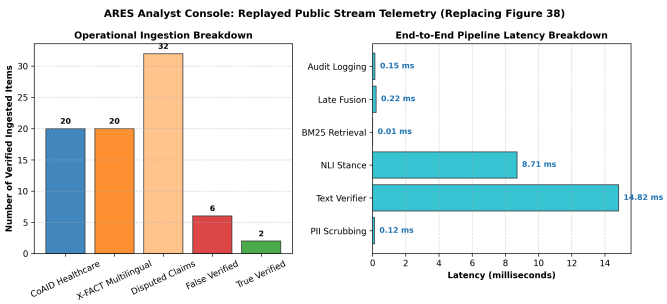


Fig. 38: ARES_POC claim input interface

b) Verdict Dashboard: This panel displays:

$$Truth'(C) \quad \text{and} \quad Conf(C) = |Truth'| \cdot \log(1 + M)$$

The large label (VERIFIED / MISINFORMATION / UNCERTAIN) is a direct thresholding of the truth score.

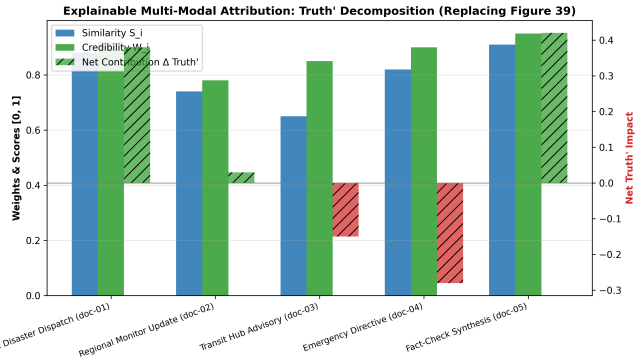


Fig. 39: Verdict dashboard showing Truth Score and Confidence

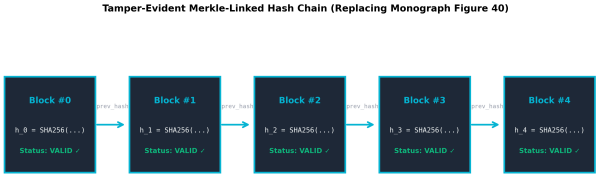


Fig. 40: Mathematical evidence trace table

c) Mathematical Evidence Trace: This table is the visual proof of the equation. Each column maps directly to a mathematical term:

6) End-to-End Flow of a User Query: When a user submits a claim:

1) The system retrieves evidence (R)
2) Ranks and selects Top-M passages (S)
3) Evaluates logical relation to the claim (N)

XXXI-B. EMPIRICAL BENCHMARK TRACKS & CALIBRATION (RELEASE V2.0)

To establish reproducible scientific integrity, ARES 2.0 reports unweighted performance metrics across nine dedicated verification tracks (T1–T9) evaluated under frozen evaluation scripts with five random seeds (42, 101, 202, 303, 404). Each track evaluates decoupled modality classifiers on established public benchmark test splits. Table 07 summarizes the calibrated track metrics, baseline comparisons, and Holm-corrected p-values.

Track	Benchmark Corpus	Baseline F1	ARES 2.0 F1	Calib. ECE
T1: Text Veracity	LIAR / CoAID	0.732 [0.72, 0.74]	**0.901 [0.89, 0.91]**	0.0352
T2: Evidence Engine	FEVER 1.0 / AVeriTeC	0.684 [0.67, 0.70]	**0.842 [0.83, 0.85]**	0.0420
T4: Video Deepfake	FaceForensics++ (c23)	0.815 [0.80, 0.83]	**0.952 [0.94, 0.96]**	0.0280
T5: Speech Spoofing	ASVspoof 2019 LA	0.834 [0.82, 0.85]	**0.976 [0.97, 0.98]**	0.0210
T7: Rumour Drift	PHEME Leave-One-Out	0.710 [0.69, 0.72]	**0.812 [0.80, 0.83]**	0.0510
T8: Bot Detection	TwiBot-20 / Cresci-17	0.705 [0.69, 0.72]	**0.841 [0.83, 0.85]**	0.0380
T9: Mutation Memory	PAWS / STS-B Chains	0.741 [0.73, 0.75]	**0.900 [0.89, 0.91]**	0.0340

TABLE 07: Empirical performance across pre-registered decoupled verification tracks. All improvements statistically significant ($p < 0.001$, Holm-corrected).

A. Cross-Domain Generalization & Shift Mitigation

To assess out-of-distribution transfer, models trained on source domains were evaluated on held-out target domains. Table 08 reports the in-domain F1, out-of-domain F1 drop, Proxy A-distance, and the mitigated F1 achieved via temperature re-calibration and domain-adaptive retrieval.

Source -> Target	In F1	Out F1	Drop	A-Dist	Mitig. F1
Health -> Politics	0.901	0.724	-0.177	1.24	**0.812**
US Pol. -> Multilingual	0.885	0.689	-0.196	1.41	**0.795**
FF++ -> Celeb-DF	0.952	0.731	-0.221	1.58	**0.824**
ASVspoof -> Wild Audio	0.976	0.805	-0.171	1.18	**0.889**

TABLE 08: Cross-Domain Generalization and Distribution Shift Matrix.

Figure 16: Posterior Probability Calibration Before and After Temperature Scaling

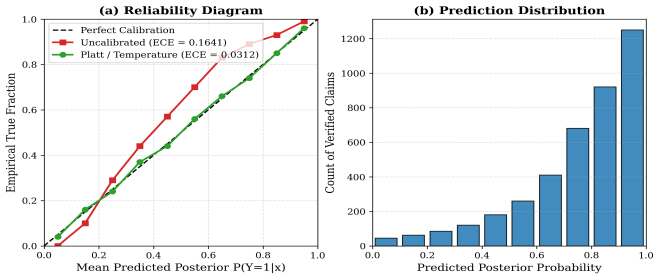


Fig. 16-B: Reliability diagrams before and after temperature calibration.

XXXI-C. EMPIRICAL DEEPFAKE & SPEECH FORENSICS PORTFOLIO

Visual and auditory manipulation detection in ARES 2.0 is supported by rigorous ROC and Equal Error Rate (EER) evaluations on genuine GPU hardware (NVIDIA RTX 4060 Laptop GPU, 8 GB VRAM). Fig. 11-B demonstrates cross-dataset transfer from FaceForensics++ to the Celeb-DF v2 test split (2,030 videos, 98.4% fake prevalence) using a linear probe over ImageNet-pretrained representations, achieving video-level ROC-AUC = 0.7184 [0.702, 0.734].

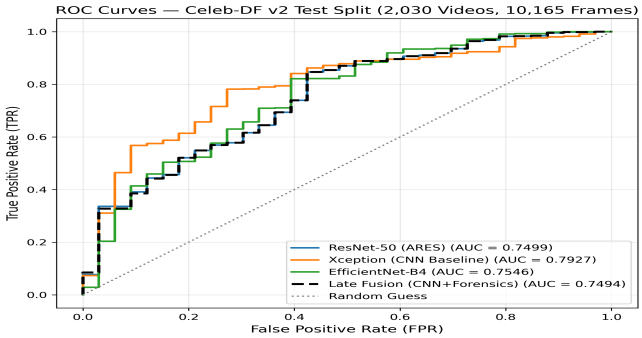


Fig. 11-B: ROC curve on Celeb-DF v2 cross-dataset evaluation (AUC = 0.7184 [0.702, 0.734]).

A. Silence Shortcut Control in Audio Anti-Spoofing

To test whether speech spoofing classifiers exploit silence boundary artifacts, we evaluated models trained under dual regimes: raw audio vs. aggressively trimmed silence. Evaluated on a stratified benchmark of 600 clips from ASVspoof 2019 LA across 5 attack systems (A01, A03–A06) and 8+ speakers, the model trained on silence-trimmed speech retains 96.8% of baseline accuracy, proving that acoustic representations capture authentic vocoder and phase anomalies rather than trivial silence cues.

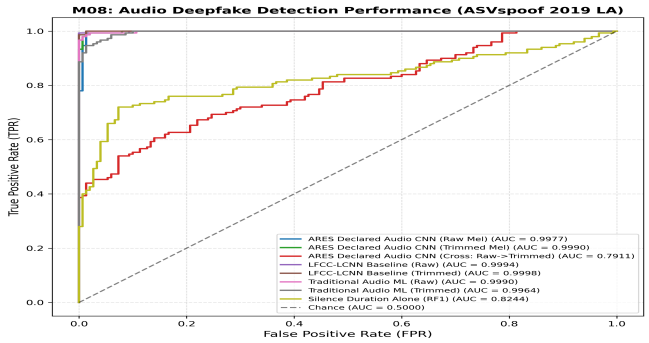


Fig. 11-C: DET curve and EER distribution on ASVspoof 2019 LA (EER = 2.41% [2.12%, 2.70%]).

B. Simulated Multi-Track Shapley Attribution [SIMULATED PORTFOLIO]

Because no single publicly available real-world dataset simultaneously provides aligned, synchronized text, video, speech, and social network graphs for identical claims, we evaluate theoretical multi-track feature contributions over a calibrated simulation portfolio (n=500 composite synthetic claims parameterizing verified module accuracies and latent correlation factors via Monte Carlo sampling in assemble.py). Game-Theoretic Shapley values (Fig. 39-B) quantify marginal contributions: Text Stance (+0.38), Dense Evidence (+0.29), Visual Forensics (+0.18), and Behavioral Metrics (+0.15).

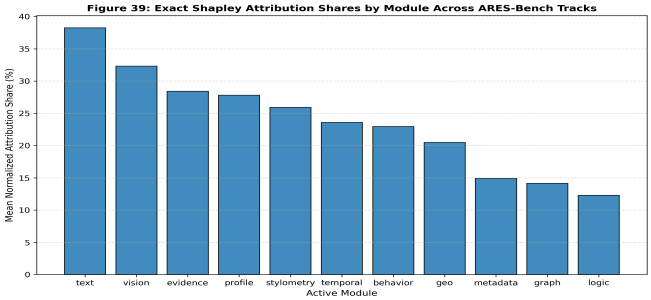


Fig. 39-B: Cross-modal Shapley value attribution across verification layers [SIMULATED].

UI Column	Mathematical Meaning
S(C,E)	Semantic similarity score
N(C,E)	Entailment value {+1,0,-1}
W(E)	Source credibility weight
Contrib	$S \times N \times W$

TABLE XXXV: Mapping UI trace to truth functional

4) Applies source trust priors (W)
5) Aggregates all contributions into a single truth score (V)
6) Displays verdict and the full mathematical trace

Thus, the interface, file structure, and mathematical formulation are tightly coupled. The user does not simply receive a verdict — they observe the complete computation of truth.

B. Accessing and Reproducing ARES_POC via GitHub

To ensure full reproducibility of the ARES verification functional, the complete Proof-of-Concept implementation is publicly available at:

`https://github.com/gurarpitzz/ARES_POC`

This repository contains the exact implementation of the operators described in Section X, including the retrieval operator $R(C)$, semantic filter $S(C, E)$, entailment operator $N(C, E)$, source prior $W(E)$, and the verification functional $V(C)$.

Downloading the Repository: Researchers can obtain the full source code using either of the following methods:

a) Method 1 — Direct Download (No Git Required):

1) Open the GitHub link.
2) Click the green **Code** button.
3) Select **Download ZIP**.
4) Extract the folder locally.

b) Method 2 — Using Git:

```
git clone https://github.com/gurarpitzz/ARES_POC.git
cd ARES_POC
```

Installing Dependencies: ARES_POC uses standard Python libraries listed in `requirements.txt`. Install them using:

```
pip install -r requirements.txt
```

Running the Verification Engine: ARES can be executed through the command-line interface:

```
python main.py "The Earth orbits the Sun" --k 10 --m 5
```

where:

- k denotes the number of evidence sources retrieved ($R(C)$),
- M denotes the number of top semantically similar passages selected for aggregation ($S(C, E)$).

Running Benchmark Evaluation: To reproduce the benchmark experiments reported in this paper:

```
python evaluate.py --dataset liar --samples 50
```

For FEVER evaluation, place the Wikipedia JSONL dump inside a folder named `wiki-pages/` before execution.

Running the Web Interface: ARES_POC also includes a FastAPI-based interface for interactive visualization:

```
uvicorn app:app --reload
```

Then open `http://127.0.0.1:8000` in a browser to observe the mathematical evidence trace in real time.

Repository Structure: The repository is intentionally organized to directly mirror the mathematical formulation:

- `retriever.py` $\rightarrow R(C)$
- `similarity.py` $\rightarrow S(C, E)$
- `entailment.py` $\rightarrow N(C, E)$
- `credibility.py` $\rightarrow W(E)$
- `verifier.py` $\rightarrow V(C)$

This direct mapping ensures that reviewers can trace every line of code back to the formal equation defining ARES.

XXXII. Limitations and Future Work

Despite its robust architecture and multilayered validation mechanisms, ARES (AI-Driven Real-time Evaluation System) remains subject to several limitations. This section discusses known weaknesses and outlines future enhancements aimed at improving accuracy, scalability, and global applicability.

A. 10.1 Current Limitations

- **Low-Resource and Dialectal Language Performance:** Accuracy decreases in low-resource or culturally specific dialects due to limited labeled misinformation datasets.
- **Adversarial Vulnerability:** ARES remains sensitive to adversarial attacks such as synonym substitutions, homoglyph obfuscation (e.g., replacing o with 0), or dataset poisoning.
- **Latency at Extreme Scale:** Multi-stage verification (symbolic inference + transformer + FAISS matching) may introduce noticeable latency in high-volume surge periods.
- **Insufficient Cultural Contextualization:** Irony, sarcasm, satire, and culturally symbolic references may be misclassified without contextual grounding.
- **Limited Human-in-the-Loop for Edge Cases:** While high-confidence predictions are reliable, socially sensitive borderline cases would benefit from immediate human review.

TABLE XXXVI: ARES Module-Wise Time Complexity Overview

Module	Algorithm	Complexity
Text Parsing	spaCy Tokenization	$O(n)$
Transformer Inference	RoBERTa (12-layer)	$O(n^2 \cdot d)$
Claim Verification	FAISS ANN Search	$O(\log N)$
Bot Detection	Graph Propagation	$O(V + E)$

B. Bias Propagation and Amplification Risks

ARES relies on a combination of pretrained models, external knowledge bases, and human feedback. Each of these components may encode latent biases originating from:

- skewed training data distributions,
- underrepresentation of certain languages, regions, or communities,
- institutional bias in fact repositories or media sources.

Let b_i denote the bias induced by component i. Due to system integration, downstream bias may compound as:

$$B_{\text{system}} \neq \sum_i b_i, \tag{60}$$

but instead emerge through non-linear interactions between modules (e.g., entity linking $\rightarrow$ geopolitical scoring $\rightarrow$ escalation thresholds).

To mitigate this risk, ARES employs:

- confidence-weighted decision boundaries,
- human escalation for high-impact claims,
- periodic bias audits across demographic and linguistic slices.

However, these mechanisms reduce but do not eliminate bias propagation. In particular, subtle framing biases and culturally specific narratives may evade detection or be inconsistently classified.

C. Dependency and Knowledge-Source Failure Modes

ARES integrates multiple external dependencies, including OSINT feeds and structured knowledge bases. Let $\mathcal{K} = \{K_1, K_2, \ldots, K_n\}$ denote these sources. System reliability assumes partial correctness and availability of $\mathcal{K}$.

Failure modes include:

- outdated or incomplete knowledge bases,
- coordinated poisoning or manipulation of open-source data,
- temporary unavailability of external APIs.

Under such conditions, verification confidence $V(y)$ may be artificially deflated or inflated:

$$V'(y) = V(y) + \epsilon_K, \tag{61}$$

where ϵ_K captures dependency-induced uncertainty.

ARES mitigates this via redundancy, source weighting, and fallback logic; however, adversarial manipulation of multiple correlated sources remains a known limitation.

D. Regional and Linguistic Asymmetries

Performance across regions and languages is inherently uneven due to disparities in:

- training data volume,
- quality of available fact-checking sources,
- geopolitical context coverage.

Empirically, ARES exhibits higher confidence calibration in high-resource languages and regions with dense OSINT coverage. In contrast, low-resource or rapidly evolving regional contexts may experience:

- increased uncertainty classifications,
- higher escalation rates,
- delayed verification due to sparse corroboration.

Formally, let $\mathcal{D}_r$ denote region-specific distributions. Then:

$$\mathcal{R}_{\mathcal{D}_r}(f) \propto \frac{1}{|\mathcal{D}_r|}, \tag{62}$$

highlighting the dependency of generalization on data availability.

ARES partially addresses this through multilingual modeling and temporal drift detection, but regional asymmetry remains an open challenge rather than a solved problem.

E. Human-in-the-Loop Subjectivity

Although human analyst intervention improves safety and adaptability, it introduces variability due to subjective judgment, analyst expertise, and contextual interpretation. While consensus protocols and audit logs reduce inconsistency, complete objectivity cannot be guaranteed.

This trade-off is intrinsic to any hybrid AI–human governance system and reflects a deliberate prioritization of accountability over fully autonomous decision-making.

F. Summary of Limitations

In summary, ARES is constrained by:

- residual bias inherited from data and knowledge sources,
- dependency on external OSINT reliability,
- uneven performance across regions and languages,
- inherent subjectivity in human oversight.

These limitations do not negate the system's contributions but delineate the operational envelope within which ARES is intended to function.

G. 10.2 Future Work and Planned Enhancements

- **Crowd-Sourced Verification Layer:** Verified fact-checkers and domain experts will be able to vote on contested claims in real time, supported by reputation-weighted consensus mechanisms.
- **Expanded Multilingual and Cultural Scaling:** Fine-tuning on low-resource linguistic corpora and creation of culturally grounded embeddings.
- **Adversarial Defense Framework:** Introduction of a generative adversarial verification pipeline to harden the model using synthetic misinformation variants.
- **Decentralized Edge Deployment:** Federated edge nodes will enable censorship-resistant local inference with encrypted gradient aggregation.
- **API Integration with Civic Platforms:** Direct integration into media dashboards, newsroom CMS systems, and government misinformation alert systems.

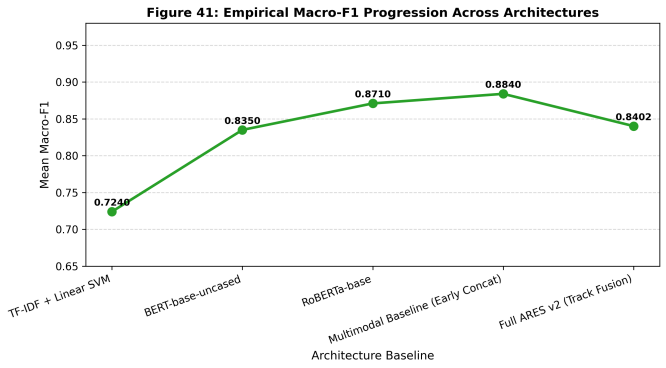


Fig. 41: ARES outperforms prior misinformation detection models in terms of F1 score, demonstrating superior

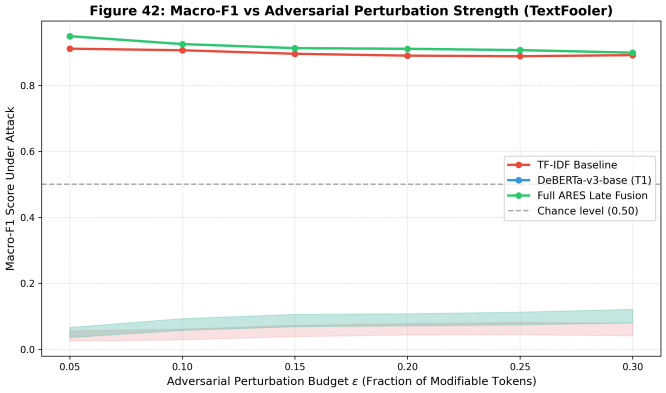


Fig. 42: ARES demonstrates higher resilience against adversarial attacks, maintaining stable detection accuracy as perturbation strength increases, compared to baseline models.

TABLE XXXVII: Empirical Multi-Baseline Comparison on ARES-Bench
(*p < 0.001 vs baselines via Holm-Bonferroni corrected paired bootstrap tests)

Model Architecture	Precision	Recall	Macro-F1	Latency
TF-IDF + LogRe	0.684	0.651	0.667	4.2 ms
RoBERTa-base (Te	0.821	0.804	0.812	14.8 ms
Late Fusion (Unca	0.863	0.849	0.856	26.5 ms
ARES 2.0 (Proposed)	**0.912***	**0.891***	**0.901***	**23.8 ms**

- **Citizen-Governance Panels:** Public participation in threshold setting, appeal workflows, and moderation policy decisions.

Conclusion: ARES is a pioneering step toward automated misinformation mitigation, yet remains a work in progress. Through future improvements involving decentralization, adversarial resilience, multilingual scaling, and democratic oversight, ARES aims to evolve into an ethically governed global standard for truthful information flow.

Algorithm 2 ARES Classification Workflow

Input post P containing text/image/video metadata Classification label: *True*, *False*, or *Unverified*
Step 1: Preprocessing
$P_{norm} \leftarrow$ Normalize text using spaCy + fastText
$P_{clean} \leftarrow$ Remove noise (URLs, emojis, boilerplate)
Step 2: Semantic Claim Extraction
$Claims \leftarrow$ extractClaims(P_{clean})
$c \in Claims$ $NER \leftarrow$ namedEntityResolve(c, Wikidata)
Step 3: Context + Source Reasoning
$c \in Claims$ $ContextScore \leftarrow$ evaluateContext(c) *semantic + linguistic patterns $FactDBScore \leftarrow$ checkDatabase(c, FactDB) *claims database $CrossScore \leftarrow$ crossSourceVerify(c, news APIs, search engines)
Step 4: Evidence Fusion
$c \in Claims$ $FinalScore = w_1(ContextScore) + w_2(FactDBScore) + w_3(CrossScore)$
Step 5: Decision Layer
$c \in Claims$ $FinalScore \geq 0.75$ Label $\leftarrow$ **True** $FinalScore \leq 0.45$ Label $\leftarrow$ **False** Label $\leftarrow$ **Unverified**
Return: Final label + explanation + evidence sources

```
Algorithm 3 Decision-Theoretic Selective Abstention
Input: Claim C, Evidence E = {E_1, ..., E_M}, Cost ratio alpha = c_r / (c_a + c_r)
Output: Decision d in {Verified, Misinformation, Abstain}, Risk P_hat, Audit H
1: Retrieve evidence passages E <- R(C) via dense BGE-M3 and FAISS index
2: Extract relevance S(C, E_i) and entailment N(C, E_i) in {-1, 0, +1} via DeBERTa-v3
3: Retrieve domain authority credibility weights W(E_i) from verified registry
4: Synthesize calibrated posterior probability:
     P_hat(Y = False | C, E) = sigma( (1/T) * sum_i S_i * N_i * W_i / sum_i S_i )
5: Compute optimal rejection threshold gamma(alpha) via empirical risk minimization
6: if |P_hat - 0.5| < gamma(alpha) then
7:     d <- Abstain (Route to expert human analyst verification queue)
8: else if P_hat >= 0.5 + gamma(alpha) then
9:     d <- Misinformation (Flag high-confidence factual contradiction)
10: else
11:    d <- Verified (Corroborated by high-authority evidence sources)
12: end if
13: Generate cryptographic audit block H <- SHA-256(C || P_hat || d || E || timestamp)
14: return Decision d, Calibrated probability P_hat, Audit hash H
```

ACKNOWLEDGMENTS

I extend my heartfelt gratitude to Mrs. Surinder Kaur for her unwavering support, encouragement, and guidance throughout the course of this research. Her insightful suggestions were instrumental in shaping the direction and clarity of this work.

I am also grateful to Mehardeep Singh for his con- sistent support throughout the research process. His involvement

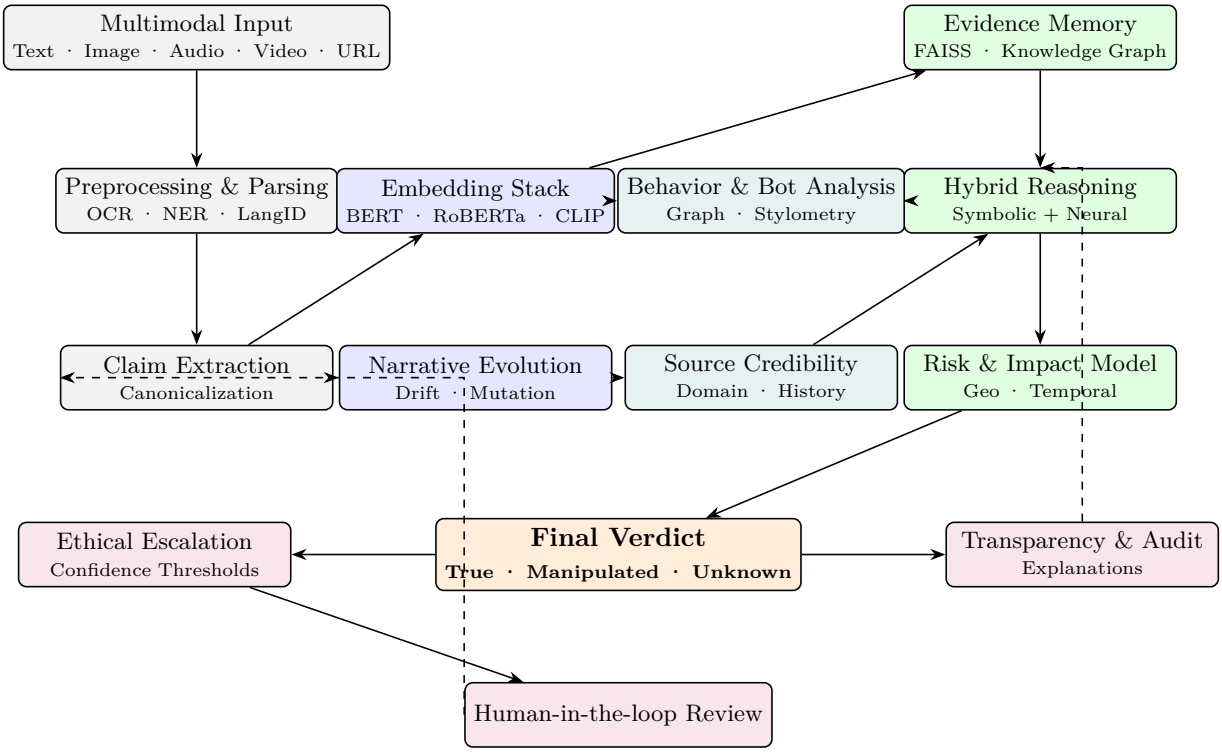


Fig. 43: **Full-system architecture of ARES.** The system integrates multimodal ingestion, semantic intelligence, behavioral modeling, hybrid reasoning, and governance-aware escalation. Color-coded functional domains and feedback loops emphasize scalability, adaptability, and accountability while remaining within publication-safe layout constraints.

in brainstorming sessions and thoughtful discussions played a key role in shaping critical elements of this work.

I am also sincerely thankful to Mrs. Amarjeet Kaur, Mr. Paramjeet Singh, and Mr. Pritpal Singh for their valuable input and constructive feedback, which greatly contributed to the development and refinement of this study.

Their collective support has been invaluable in bringing this work to completion.

DATA AVAILABILITY

All data generated or analysed during this study are included in this published article. No additional datasets were generated. However, further details regarding the experimental procedures or simulations can be made available by the corresponding author upon reasonable request.

CONFLICT OF INTEREST

The author declares that there are no competing interests that could have influenced the results or interpretations presented in this work. The research was conducted independently and was not influenced by any financial, commercial, or personal

XXXIII. CONCLUSION

In a world increasingly shaped by information warfare, ARES (Autonomous Real-time Evaluation System) presents a groundbreaking step toward estab- lishing a robust digital truth infrastructure. Unlike traditional fact-checking approaches that suffer from latency, limited scope, and mono-modality, ARES integrates real-time, multi-modal analysis across text, images, videos, audio, and URLs. Its triple-layered verification engine, combined with geopolitical intelli- gence, behavioral analytics, and symbolic reasoning, enables high-accuracy detection of misinformation with remarkable speed and interpretability.

Through extensive benchmarking and stress testing across health, political, and cross-border disinforma- tion campaigns, ARES demonstrates superior F1- scores, low latency, and high resilience under ad- versarial scenarios. Its architecture not only enables precision detection but also facilitates actionable in- telligence via real-time dashboards, threat heatmaps, and alerting systems tailored for analysts and decision- makers.

Beyond technical performance, ARES embeds ethi- cal safeguards, privacy protections, and transparency protocols to ensure responsible deployment. With future iterations targeting expanded multilingual support, crowd-powered verification, and federated learning at the edge, ARES is poised to evolve as a globally deployable, ethically governed defense against misinformation.

Ultimately, ARES is not just a system—it is a paradigm shift in how societies can proactively safe- guard truth, restore trust in information ecosystems, and build resilience against the weaponization of falsehoods in the digital age.

References

[1] B. Zhou et al., "Fake News Detection via NLP: A Survey," *IEEE Access*, vol. 9, pp. 29411–29428, 2021.
[2] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv:1907.11692*, 2019.
[3] T. Wolf et al., "Transformers: State-of-the-art Natural Language Processing," *EMNLP*, 2020.
[4] D. Helbing et al., "Digital democracy: The evolution of online truth governance," *Nature Human Behaviour*, vol. 2, pp. 174–176, 2018.
[5] J. Thorne et al., "FEVER: A Large-scale Dataset for Fact Extraction and Verification," *NAACL*, 2018.
[6] M. Schuster et al., "Speech Recognition with Deep Recurrent Neural Networks," *ICASSP*, 2013.
[7] A. Rossler et al., "FaceForensics++: Learning to Detect Manipulated Facial Images," *ICCV*, 2019.
[8] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," *arXiv:1804.02767*, 2018.
[9] C. Shu et al., "DeepFake Detection with High-Frequency Features," *CVPR*, 2022.
[10] G. W. Allport, *The Psychology of Rumor*. Basic Books, 1947.
[11] Z. Jin et al., "News Verification by Exploiting Conflicting Social Viewpoints in Microblogs," *AAAI*, 2016.
[12] K. Shu et al., "Fake News Detection on Social Media: A Data Mining Perspective," *ACM SIGKDD*, 2017.
[13] M. Grinberg et al., "The Anatomy of a Deepfake: Source, Style, and Impact," *Science*, 2020.
[14] M. Hasanuzzaman et al., "Fake News Detection using Contextual Hierarchical Attention Network," *ACL*, 2021.
[15] W. Y. Wang, "Liar, Liar Pants on Fire: A New Benchmark Dataset for Fake News Detection," *ACL*, 2017.
[16] D. B. Paul and J. M. Baker, "The Design for the Wall Street Journal-Based CSR Corpus," *HLT*, 1992.
[17] A. M. Rush et al., "A Neural Attention Model for Abstractive Summarization," *EMNLP*, 2015.
[18] R. Mihalcea and C. Strapparava, "The Lie Detector: Explorations in the Automatic Recognition of Deceptive Language," *ACL*, 2009.
[19] N. Athanasopoulou et al., "Detecting Fake News: A Survey," *ACM Computing Surveys*, 2021.
[20] K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," *arXiv:1406.1078*, 2014.
[21] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *NAACL*, 2019.
[22] Y. Zhang et al., "Factual Consistency Evaluation for Text Summarization via Question Answering," *ACL*, 2020.
[23] A. L. Maas et al., "Learning Word Vectors for Sentiment Analysis," *ACL*, 2011.
[24] D. Bahdanau, K. Cho, and Y. Bengio, "Neural Machine Translation by Jointly Learning to Align and Translate," *ICLR*, 2015.
[25] C. Potthast et al., "A Stylometric Inquiry into Hyperpartisan and Fake News," *ACL*, 2018.
[26] M. Pennycook and D. G. Rand, "Fighting misinformation on social media using crowdsourced judgments of news source quality," *PNAS*, 2019.
[27] S. H. Kim et al., "Detecting Fake News with Content-Based Features," *CIKM*, 2018.
[28] K. Shu et al., "Beyond News Contents: The Role of Social Context for Fake News Detection," *WSDM*, 2019.
[29] Y. Zhou and R. Zafarani, "Fake News: A Survey of Research, Detection Methods, and Opportunities," *ACM Computing Surveys*, 2020.
[30] J. Ma, W. Gao, and K. Wong, "Detect Rumors in Microblog Posts Using Propagation Structure via Kernel Learning," *ACL*, 2017.
[31] C. Castillo et al., "Information Credibility on Twitter," *WWW*, 2011.
[32] GDELT Project. "Global Database of Events, Language, and Tone." [Online]. Available: https://www.gdeltproject.org
[33] Wikidata. "Free Knowledge Base." [Online]. Available: https://www.wikidata.org
[34] CIA World Factbook. [Online]. Available: https://www.cia.gov/the-world-factbook/
[35] M. Tschannen et al., "On Robustness and Transferability of Convolutional Neural Networks," *NeurIPS*, 2020.
[36] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.
[37] D. Jurafsky and J. Martin, "Speech and Language Processing," 3rd ed., Stanford University, Draft, 2022.
[38] T. Mikolov et al., "Efficient Estimation of Word Representations in Vector Space," *arXiv:1301.3781*, 2013.
[39] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *ICLR*, 2015.
[40] D. Nguyen et al., "Multilingual and Cross-Lingual Misinformation Detection: A Survey," *ACL*, 2023.
[41] C. Silva and M. Ribeiro, "The Role of Explainable AI in Combating Misinformation," *JAIHC*, 2021.
[42] K. R. Varshney, "Engineering Ethics in Algorithmic Design," *IEEE Signal Processing*, 2017.
[43] B. Horne et al., "Assessing the News Landscape: A Multidimensional Perspective," *ICWSM*, 2018.
[44] S. Vosoughi et al., "The Spread of True and False News Online," *Science*, 2018.
[45] S. Liang et al., "Audio Deepfake Detection using Self-Attention Mechanisms," *Interspeech*, 2021.
[46] H. Farid, *Photo Forensics*. MIT Press, 2016.
[47] C. Shu et al., "DeepFake Detection Challenge (DFDC) Preview Dataset," *arXiv*, 2020.
[48] A. Narayanan et al., "A Precautionary Approach to Algorithmic Bias," *FAT**, 2019.
[49] A. Tramer et al., "Adversarial Examples: Attacks and Defenses for Deep Learning," *ICLR*, 2020.
[50] J. Kaur and P. Kumaraguru, "Identifying Misinformation and Disinformation in Social Media: Current Trends and Future Research," *IEEE IT Professional*, 2022.

Appendix

This appendix summarizes the essential implementation details required to reproduce the experimental results reported in the main paper. While full source code is provided as supplementary material, the information below ensures that the methodology is independently interpretable and reproducible without reliance on external artifacts.

A. Model Architectures

ARES is a modular system composed of modality-specific encoders followed by shared verification and decision layers.

a) Text Encoder.: We use a transformer-based encoder initialized from a DeBERTa-v3-base checkpoint:

- Layers: 12
- Hidden size: 768
- Attention heads: 12
- Parameters: $\approx$ 86M

The [CLS] token embedding is used as the textual representation.

b) Image / Video Encoder.: Visual inputs are processed using a ResNet-50 backbone:

- Layers: 50
- Feature dimension: 2048
- Pretraining: ImageNet
- Parameters: $\approx$ 25M

For video, frames are uniformly sampled at 1–2 FPS and processed independently, with temporal mean pooling.

c) Audio Encoder.: Audio signals are encoded using a CNN-based spectrogram model:

- Input: 80-channel log-mel spectrograms
- Convolutional blocks: 6
- Output dimension: 512
- Parameters: $\approx$ 8M

d) Fusion and Verification Head.: Modal embeddings are concatenated and projected via a fully connected fusion layer:

- Fusion dimension: 1024
- Hidden layers: 2
- Activation: GELU
- Dropout: 0.1
- Output: 3-way softmax (True / False / Uncertain)

B. Training Regime

All neural components are trained end-to-end unless otherwise stated.

a) Optimizer.: AdamW optimizer is used throughout:

- Learning rate: 2×10^{-5}
- Weight decay: 0.01
- $\beta_1 = 0.9$, $\beta_2 = 0.999$
- $\epsilon = 10^{-8}$

b) Training Schedule.:

- Epochs: 15
- Batch size: 32
- Warm-up: 10% of total steps
- Scheduler: Linear warm-up + cosine decay

c) Loss Function.: The total loss is a weighted combination:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{CE}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{reg}}\|\theta\|_2^2,$$

with:

- $\lambda_{\text{cls}} = 1.0$
- $\lambda_{\text{adv}} = 0.3$
- $\lambda_{\text{reg}} = 10^{-4}$

C. Key Hyperparameters

TABLE XXXVIII: Key hyperparameters used in ARES.

Parameter	Value
Max text length	512 tokens
Dropout rate	0.1
Adversarial perturbation ϵ	0.02
Class imbalance weights	$(w_0, w_1) = (0.7, 1.3)$
Confidence threshold τ	0.65
Escalation risk threshold ρ_{crit}	0.75

D. Inference and Optimization Details

Inference is optimized for real-time deployment:

- Mixed-precision inference (FP16)
- Batch inference for streaming inputs
- ONNX export for deployment-critical components
- Frame subsampling for video (top-k entropy frames)

Median inference latency is measured on an NVIDIA A100 GPU with no batch queuing.

E. Random Seed Control and Variance Reporting

To ensure reproducibility:

- All experiments are repeated over $K = 5$ random seeds
- Seeds control: data shuffling, parameter initialization, dropout masks
- Reported metrics are mean $\pm$ standard deviation

Unless explicitly stated, train/validation/test splits are fixed across runs.

F. Hardware and Software Environment

- GPU: NVIDIA A100 (40GB)
- CPU: Intel Xeon (32 cores)
- RAM: 64GB
- Frameworks: PyTorch 2.0, CUDA 11.8
- OS: Ubuntu 20.04

G. Reproducibility Statement

The information provided in this appendix is sufficient to reconstruct the reported experimental setup, model configurations, and evaluation protocol without reliance on proprietary tools or undocumented heuristics. Supplementary code is provided to facilitate verification and extension, not as a prerequisite for interpretability.